More Stores or Bigger Stores? Cross-Border Demand and Retail Entry in New Mexico's Cannabis Market

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Abstract

When a retail market gains customers, who captures the additional demand: existing stores or new entrants? The answer matters for competition, market structure, and the rents earned by incumbents, yet separating these responses is empirically difficult. We study New Mexico's recreational cannabis market, which opened in April 2022 without a statewide cap on retail licenses. Dispensaries near Texas, Oklahoma, and Utah could attract consumers from states without legal recreational sales, while those near Colorado and Arizona faced consumers with legal alternatives at home. Using storefront-level sales and ownership records from 2022 to 2026, we find that markets exposed to captive cross-border demand generate 2.4 times as much revenue and support nearly twice as many dispensaries, conditional on resident population. Greater store density accounts for roughly three-quarters of the revenue gap, while the revenue-per-store premium is imprecisely estimated. Chains initially appear to capture this opportunity first, but their advantage largely reflects medical dispensaries already operating when recreational sales began. Among subsequent openings, chains do not respond more strongly than independents. An estimated model of entry and market adjustment reproduces these patterns and highlights the role of inherited capacity in accelerating market formation. The findings suggest that when entry is relatively open, additional demand is accommodated primarily through new retail capacity rather than persistently higher sales at existing stores.

Keywords: Market entry · Retail market structure · Cross-border demand · Incumbency · Entry regulation · Cannabis
JEL Classification: L11 · L13 · L51 · L81 · R12

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1 Introduction

Larger markets support more firms. This is one of the central predictions of entry theory (Bresnahan & Reiss, 1991), yet when a market gains customers it is far from obvious who benefits. Existing retailers may capture the additional spending through higher sales, or new firms may enter and compete for it. These two responses have different implications for market structure, competition, and the rents earned by incumbents. Spatial competition models make the distinction particularly clear: when demand grows by extending a market's geographic reach, free entry can support more outlets while leaving sales per outlet largely unchanged; when demand becomes denser within an existing market, sales per outlet can rise even with free entry (Salop, 1979). The challenge is to determine which margin absorbs a demand increase in practice. Without separating entry from incumbent sales, a larger retail market tells us surprisingly little about how competition responds to growth.

This distinction becomes even more important when firms cannot respond immediately. Opening a store requires premises, financing, regulatory approvals, and time. Firms that already possess these resources may capture a new market before other entrants are ready, even if they have no lasting advantage once entry catches up. An early lead by large retail chains could therefore reflect the capacity they inherited from an earlier market rather than an intrinsic advantage of chain organization. Understanding the response to demand growth requires answering two related questions: how much of the new demand is absorbed through entry, and which firms are able to enter first? These questions matter for interpreting the relationship between market size and concentration, the role of entry barriers, and the sources of first-mover advantage.

New Mexico's legalization of recreational cannabis offers an opportunity to examine both questions. Adult-use sales began on April 1, 2022, under a law that imposes no statewide cap on retail licenses and prevents local governments from prohibiting licensed dispensaries (State of New Mexico, 2021). But the demand available to retailers differed sharply across the state's borders. Texas, Oklahoma, and Utah did not permit recreational cannabis sales during the sample, while Colorado and Arizona already had legal recreational markets. A dispensary near Texas could therefore attract consumers who had no legal recreational retailer in their home state, while a dispensary near Colorado faced consumers with access to legal retailers closer to home. The result was a geo-

graphically concentrated source of additional demand within a single state and regulatory system. Because New Mexico's administrative records report monthly sales for every licensed storefront, we can follow the number of stores, their revenues, and their operators from the market's opening in 2022Q2 through 2026Q2.

We use this setting to study how captive cross-border demand shapes retail market structure. Our main comparison is between ZIP codes near states without legal recreational sales and ZIP codes near states where recreational sales were already legal. Both groups are located along state borders, but they differ in the legal purchasing options available to consumers on the other side. This comparison does not rely on a conventional difference-in-differences design: recreational sales began statewide at the same time, and the administrative sales panel has no pre-legalization period. Instead, we estimate conditional differences across border markets and examine how those differences evolve after legalization. The identifying assumption is that, after accounting for observed local characteristics, the groups would have had comparable retail outcomes in the absence of differential cross-border demand. We probe this assumption using predetermined controls, an imputation estimator fitted on unexposed ZIP codes, exposure measures based on distance and road-network travel time, and placebo outcomes in non-cannabis retail, where data from 2018–2021 provide a pre-period.

Our first finding is that captive demand supports substantially larger cannabis markets, but most of the difference appears in the number of stores rather than sales per store. Conditional on resident population, ZIP codes exposed to neighboring states without recreational sales generate 2.44 times as much cannabis revenue as those near states with legal recreational sales (95 percent confidence interval 1.41–4.23). They also host 1.94 times as many dispensaries. The revenue ratio remains between 2.2 and 2.6 in every quarter, suggesting that the cross-border demand advantage persisted as the market developed. An exact decomposition of the revenue-per-resident gap attributes 73 percent to differences in stores per resident and 27 percent to differences in revenue per store. The latter estimate is imprecise: its 95 percent confidence interval ranges from -0.16 to 0.68 log points. We therefore cannot rule out a meaningful sales premium for individual stores. Nevertheless, the evidence is inconsistent with a market in which existing outlets absorb all the additional demand. Entry accounts for the larger share of the observed difference.

Several features of the data reinforce this interpretation. The revenue advantage comes primarily

from recreational sales, which are available to nonresident consumers, rather than medical sales, which require a New Mexico medical cannabis card. Stores already operating when recreational sales began did not initially earn more in markets with greater cross-border exposure, nor did their sales subsequently grow faster than those of stores elsewhere. Instead, their revenues decline as additional competitors open nearby. Moreover, dispensary density increases with the number of residents of neighboring states without legal recreational sales who live within reach of a market. This relationship appears whether exposure is measured using geographic distance or driving time. Revenue per store shows no comparably clear exposure gradient. Together, these findings suggest that additional customers are served largely by expanding retail capacity rather than by persistently increasing sales at incumbent outlets.

But if entry absorbs most of the demand, who supplies the new stores? At first, the market appears to favor chains. When operators are classified as chains if they operate at least two New Mexico storefronts at any point during the sample, exposed ZIP codes begin with substantially more chain-operated stores relative to comparison markets. In 2022Q2, the difference is approximately 26 chain-associated operators per 100 ZIP codes, compared with about 6 independents. This apparent advantage, however, disappears within a year. Its origin lies largely in stores already operating when recreational sales began. Of the 78 storefronts initially present in ZIP codes near nonlegal-state borders, 61 belonged to operators classified as chains over the sample, and 45 of those were medical dispensaries that began selling recreational cannabis immediately. Statewide, 83 percent of medical-incumbent storefronts belonged to operators that already ran multiple stores.

The distinction between inherited capacity and new entry changes the interpretation of the chain advantage. Once stores operating at legalization are excluded, chains do not show a consistently stronger response to captive demand under alternative classifications of organizational type. Openings increase substantially for both established operators and firms entering with their first storefront. What distinguishes early entrants is not simply the number of stores they operate, but whether they participated in the medical cannabis market before legalization. Conditional on operator size, medical incumbents opened subsequent stores an average of 3.2 quarters earlier than other operators. This is a conditional association rather than a causal estimate of experience, but it points to an important mechanism: access to existing premises, regulatory knowledge, and other resources accumulated before legalization may determine who can respond first, even when those

advantages do not translate into a lasting difference in entry rates.

These findings motivate a framework that separates the size of a market from the speed at which retail capacity adjusts. Captive demand increases the number of stores a market can support, while preparation completed before legalization affects how quickly potential entrants can open. Firms with existing medical operations may begin with more of the necessary preparation already completed, whereas other entrants must establish their operations after the recreational market opens. We estimate an empirical model combining Poisson entry equations for first-store openings and expansions by different types of established operators, a store-exit equation, and a revenue equation linking total market sales to the number of dispensaries. Estimated using the first nine quarters, the model predicts the stock of stores in each exposure group in 2026Q2 within approximately 20 percent and reproduces the earlier expansion of medical incumbents. The estimated elasticity of market revenue with respect to the number of stores is 0.81 (SE 0.10). Thus, additional stores are associated with higher total market sales, but the increase is less than proportional, consistent with competition reducing the sales available to existing retailers.

The model also helps distinguish the consequences of inherited retail capacity from those of faster entry. Removing the 226 medical storefronts inherited at legalization reduces first-year store-quarters by 19 to 64 percent across the counterfactual bounds. The range reflects uncertainty about how quickly other entrants would have replaced those stores, a response that the observed data cannot identify directly. A separate exercise that shortens the entry pipeline by one quarter increases store-quarters over the sample by approximately 5 to 6 percent. These exercises suggest that the capacity available when a market opens can have substantial consequences for its initial development, even if its effect on long-run market structure is harder to establish.

The paper contributes to three strands of research. First, it connects empirical work on entry and market structure (S. Berry & Reiss, 2007; Bresnahan & Reiss, 1991; Coccorese & Pellecchia, 2022; Dunne, Klimek, Roberts, & Xu, 2013; Mazzeo, 2002; Seim, 2006) with the distinction between geographic expansion and growth in demand density (Campbell & Hopenhayn, 2005; Münter, 2023; Salop, 1979). Rather than inferring the consequences of market growth from firm counts alone, we observe revenue and entry jointly and measure how additional demand is divided between the two margins. An exact decomposition accommodates markets without operating stores and makes the uncertainty surrounding the revenue-per-store margin explicit. The results provide evidence of a

setting in which geographically concentrated demand growth is associated primarily with greater retail density, as predicted by spatial models with entry.

Second, the paper speaks to the literature on retail chains, organizational advantage, and entry dynamics (Hollenbeck, 2017; Hollenbeck & Giroldo, 2022; Holmes, 2011; Jia, 2008; Talamas Marcos, 2025). An early concentration of chain-operated stores might appear to indicate that multi-store firms expand more quickly when new markets become available. Our findings reveal a different explanation: much of the apparent head start was inherited from the medical market. This distinction is difficult to see without observing ownership, preexisting establishments, and the timing of new openings. It connects the entry literature to research on the importance of pre-entry experience, accumulated capabilities, and first-mover advantages (Helfat & Lieberman, 2002; Klepper & Simons, 2000; Lieberman & Montgomery, 1988). More broadly, it shows why organizational classifications that use information from future expansion can obscure the difference between being prepared to enter and expanding rapidly after entry becomes possible.

Third, the paper adds a supply-side perspective to research on cross-border shopping and cannabis legalization (Asplund, Friberg, & Wilander, 2007; Chandra, Head, & Tappata, 2014; Hansen, Miller, & Weber, 2020; Hao & Cowan, 2020; Hollenbeck & Uetake, 2021; Lovenheim, 2008). Cross-border demand does more than redirect consumer spending. It can change where stores open, how quickly retail networks form, and how much of the resulting market opportunity is captured by incumbents rather than new entrants. A descriptive comparison with Colorado, where local governments may restrict cannabis retail entry, offers suggestive evidence that this adjustment may look different when new stores face additional constraints. The broader lesson is that the effect of a demand increase on incumbent revenues depends not only on the size of the opportunity, but also on how readily firms can enter and how much capacity already exists when the opportunity arises.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting and data. Section 3 develops the economic framework and its predictions, and Section 4 presents the empirical strategy. Section 5 examines the market-level response to captive demand, while Section 6 studies organizational differences in entry and their timing. Section 7 presents the estimated model and its validation. Section 8 reports the counterfactual exercises and the descriptive comparison with Colorado. Section 9 concludes. Online Resource 1 contains additional tables and figures.

2 Institutional Background and Data

2.1 Institutional setting

New Mexico's Cannabis Regulation Act was enacted in 2021, and adult-use retail sales began on April 1, 2022. The Act does not limit the number of retail licenses the Cannabis Control Division may issue and prevents municipalities and counties from prohibiting licensed cannabis businesses. Local governments may regulate the time, place, and manner of operation through zoning and operating hours, but they may not require separation from schools or day-care centers greater than 300 feet (State of New Mexico, 2021). Opening a store still requires a license, a compliant location, and inspections, so entry takes time, but no numerical cap binds.

The market did not open with a blank slate. Medical cannabis had been legal in New Mexico for more than a decade, and medical licensees could begin adult-use sales at their existing locations on the first day. Of the storefronts in the data, 304 report sales in 2022Q2, and 226 of them report medical sales in April 2022. These are referred to as medical incumbents.

New Mexico borders five states. Texas, Oklahoma, and Utah did not permit recreational sales during the sample. Colorado has allowed adult-use sales since 2014 and Arizona since 2021. Colorado also provides a contrasting regulatory regime: Amendment 64 lets local governments prohibit retail marijuana establishments (State of Colorado, 2012), and in 2019 fewer than a quarter of Colorado counties allowed all types of recreational facilities (Payán, Brown, & Song, 2021). Colorado is used only for a descriptive comparison.

2.2 Sales, storefronts, and operators

The primary source is the Cannabis Control Division's reporting portal (New Mexico Cannabis Control Division, 2026), which publishes monthly gross medical and adult-use sales for each licensee at each reported address. Revenue is total gross sales, aggregated to quarters. Reported sales over 2022Q2–2026Q2 total $2.31 billion statewide, of which 27 percent are medical; in the 209 border ZIP codes of the main sample the medical share is 18 percent. The records report revenue but not prices or quantities, so revenue per store combines both.

A storefront is a unique licensee–address pair. Geocoding the addresses with the Census Bureau's batch geocoder yields 1,255 storefronts, of which 1,232 report positive sales. A storefront

is active in a quarter if it reports positive sales in that quarter, and each storefront is assigned to the ZIP code of its address. Because the records cover every licensee in every month, a ZIP code without reported sales has no operating dispensary rather than a missing value.

Storefronts are linked to 652 operators using licensee and trade names. Organizational form can be defined in several ways, and the choice matters for comparisons over time. The paper uses three definitions. Under the *lifetime* definition, an operator is a chain if it runs at least two New Mexico storefronts at any point in the sample; 181 operators are chains, and they account for 784 storefronts. This definition uses information from the end of the sample. Under the definition *at market opening*, an operator is a chain if it runs at least two storefronts in 2022Q2; the label is fixed before any post-legalization entry. Under the *status-at-entry* definition, each storefront is classified when it first reports sales: it is an established-operator entry if its operator already runs another New Mexico storefront and a new-operator entry otherwise. Of the 1,232 active storefronts, 370 are established-operator entries, 395 are first locations of operators that later open a second store, and 467 belong to operators that never do.

2.3 Geography and border exposure

ZIP-code populations and centroids come from the SimpleMaps U.S. ZIP database (SimpleMaps, 2026), which also covers neighboring states, and residents within 1 and 2 kilometers of each dispensary come from 2020 Census blocks (U.S. Census Bureau, 2021). Exposure is defined from geography alone, so it does not depend on where firms chose to open. A ZIP code is exposed to a neighboring state if its centroid lies within 75 miles of that state's border and at least 1,000 residents of that state live within 75 miles of the centroid. This rule divides New Mexico's 368 ZIP codes into 92 *nonlegal-border* ZIP codes exposed to Texas, Oklahoma, or Utah; 117 *always-legal-border* ZIP codes exposed to Colorado or Arizona but not to a nonlegal state; and 159 *interior* ZIP codes, including the Albuquerque metropolitan area. Fourteen ZIP codes exposed to both kinds of border are assigned to the nonlegal-border group.

Exposure is also measured continuously. The baseline measure, E_z, counts nonlegal-state residents within 75 miles of the ZIP-code centroid; alternatives use 50- and 100-mile radii and distance-decay weights. A new measure uses road-network travel times from each New Mexico ZIP-code centroid to the centroid of every populated Texas, Oklahoma, and Utah ZIP code within 150 miles,

computed with the Open Source Routing Machine on OpenStreetMap roads (Luxen & Vetter, 2011; Project OSRM, 2026); it counts nonlegal-state residents within 60, 90, or 120 minutes of driving. These are modeled travel times, not observed trips, and all exposure measures capture potential customers rather than observed cross-border purchases, which the data do not record.

Figure 1 maps the three groups. Pooling all quarters, nonlegal-border ZIP codes have 0.41 dispensaries per 1,000 residents and $95 of quarterly revenue per resident, compared with 0.19 dispensaries and $34 in always-legal-border ZIP codes and 0.28 dispensaries and $60 in interior ZIP codes.

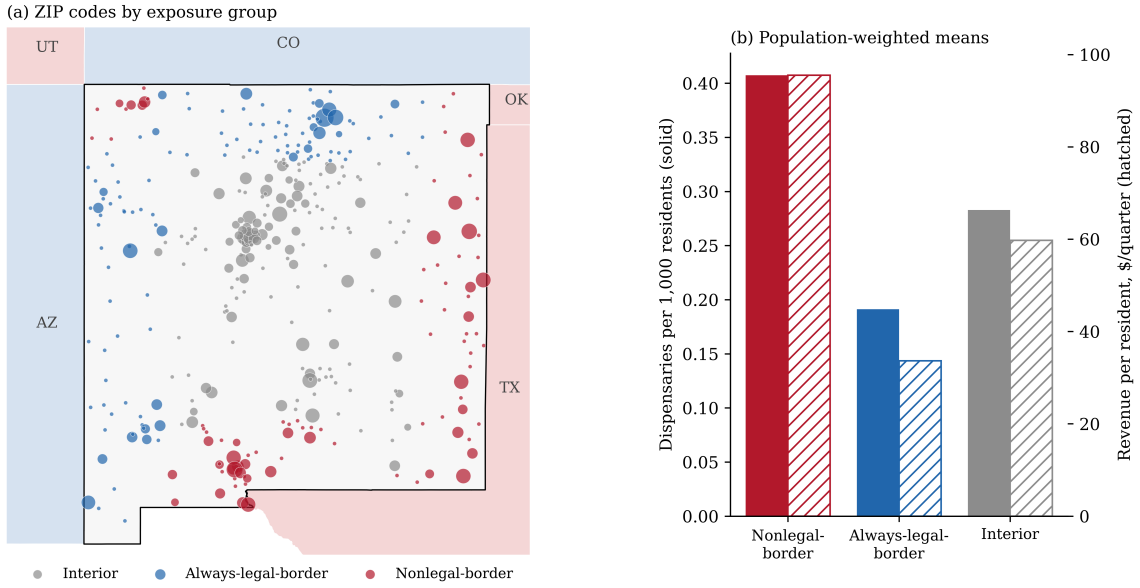


Fig. 1 Border exposure and dispensary density in New Mexico

Notes: Panel (a): ZIP-code centroids by exposure group; marker area increases with average dispensaries per 1,000 residents over 2022Q2–2026Q2. Neighboring states are shaded according to whether they had an adult-use market during the sample (Texas, Oklahoma, Utah: no; Colorado, Arizona: yes). Panel (b): dispensaries per 1,000 residents (solid bars, left axis) and quarterly revenue per resident (hatched bars, right axis), computed as group totals divided by group population over all ZIP-quarters.

2.4 Samples and outcomes

The main sample is a balanced panel of the 209 nonlegal- and always-legal-border ZIP codes over 17 quarters, 2022Q2–2026Q2, or 3,553 ZIP-quarters in 23 counties. The outcomes are quarterly revenue per resident, dispensaries per 1,000 residents, and revenue per dispensary, which is defined for ZIP-quarters with at least one dispensary. The first two enter regressions through the inverse

hyperbolic sine, which accommodates zeros; revenue per dispensary enters in logs. Because inverse-hyperbolic-sine coefficients depend on the units of the outcome and mix extensive and intensive margins (Bellemare & Wichman, 2020; Chen & Roth, 2024), they are not converted into percentage effects. Magnitudes come from Poisson pseudo-maximum-likelihood regressions (Santos Silva & Tenreyro, 2006), from regressions in levels, and from an exact decomposition in logs of group totals. Table 1 reports summary statistics.

The dispensary panel contains the 1,232 storefronts that report positive sales, observed in the quarters in which they operate (10,653 storefront-quarters). Three alternative samples serve as robustness checks: a pooled-control sample that adds the 159 interior ZIP codes to the comparison group, a sample of ZIP-quarters with at least one dispensary, and a panel of 78 town markets (Online Resource 1, Table OR1). The entry model in Section 7 uses all 368 ZIP codes. The placebo exercise uses establishment counts in grocery stores and total retail trade from the ZIP Codes Business Patterns for 2018–2023 (U.S. Census Bureau, 2025), which include four years before legalization.

Table 1: Summary statistics, ZIP-quarter panel

	Nonlegal-border (1)	Always-legal-border (2)	Difference (1)−(2) (3)	Interior (4)
Panel A: Outcomes				
Revenue per resident ($/quarter)	60.5	11.9	48.6***	23.2
	[220.1]	[48.3]	(17.0)	[44.6]
Dispensaries per 1,000 residents	0.277	0.145	0.132*	0.178
	[0.646]	[0.852]	(0.076)	[0.362]
Revenue per dispensary ($000/quarter)[a]	199.2	126.6	72.6***	195.3
	[175.3]	[92.0]	(20.5)	[159.1]
Independent operators (count)	0.99	0.15	0.84***	0.75
	[2.31]	[0.56]	(0.18)	[1.91]
Chain operators (count)	1.24	0.20	1.05***	1.46
	[2.92]	[0.91]	(0.22)	[3.17]
Panel B: Market activity				
Dispensaries (count)	2.32	0.36	1.96***	2.34
	[5.26]	[1.45]	(0.41)	[5.17]
Any dispensary (share of ZIP-quarters)	0.315	0.132	0.183***	0.366
	[0.465]	[0.338]	(0.052)	[0.482]
Total revenue ($000/quarter)	543.2	63.1	480.1***	493.6
	[1,679.2]	[322.7]	(125.8)	[1,151.5]
Market age (quarters since first dispensary)[a]	7.3	6.4	0.9**	7.6
	[4.8]	[4.6]	(0.4)	[4.8]
Panel C: Market characteristics (time-invariant)				
Resident population	5,688	1,875	3,812***	8,262
	[10,657]	[3,738]	(1,009)	[14,943]
Nonlegal-state residents within 75 mi (000s)	288.2	0.0	288.1**	0.1
	[325.9]	[0.1]	(131.8)	[0.2]
Legal-state residents within 75 mi (000s)	11.0	30.8	−19.8**	0.0
	[29.9]	[23.8]	(8.9)	[0.1]
Distance to nearest nonlegal-state line (mi)	29.7	124.8	−95.1***	138.3
	[18.3]	[30.1]	(9.4)	[42.5]
Distance to nearest legal-state line (mi)	150.3	36.0	114.2***	130.3
	[82.1]	[18.3]	(25.2)	[51.3]
ZIP codes	92	117		159
ZIP codes with a dispensary in any quarter	35	24		66
ZIP-quarters	1,564	1,989		2,703
ZIP-quarters with ≥1 dispensary	493	262		990
Counties	13	13		19

Notes: ZIP-quarter observations, 2022Q2–2026Q2. Columns (1), (2), and (4) report means with standard deviations in brackets. Column (3) reports the coefficient from a regression of the variable on a nonlegal-border indicator in the main sample, without controls; standard errors clustered by county in parentheses. Revenue includes medical and adult-use sales. Operator counts use the lifetime classification. [a]Defined for ZIP-quarters with at least one dispensary. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

3 Framework: Demand, Entry, and Inherited Capacity

The framework has two parts. The first asks how many stores a market supports once entry has run its course and how additional demand divides between the number of stores and sales per store. The second asks how quickly the market gets there and which firms arrive first. Section 7 turns both parts into an estimable model.

3.1 Demand and the store margin

Consider the circular-city model of Salop (1979). Consumers are spread uniformly along a circle of length ℓ with density ρ; each buys one unit from the store that minimizes price plus travel cost t per unit of distance; stores have marginal cost c and fixed cost f; and firms enter until profits are zero. Total demand is $S = \rho\ell$. For a nonlegal-border market, S includes residents of the neighboring state who are willing to travel. With n symmetric stores, the price equilibrium is $p = c + t\ell/n$ and each store serves $q = \rho\ell/n$ customers. Free entry sets $(p - c)q = f$, so

$$\bar{q} = \sqrt{\rho f/t}, \qquad \bar{p} = c + \sqrt{tf/\rho}, \qquad n^* = S/\bar{q}, \tag{1}$$

and revenue per store is $\bar{p}\bar{q} = c\bar{q} + f$.

Additional demand can arrive in two ways. If out-of-state customers extend the area a market serves while the density of customers along it changes little, S rises through ℓ: the number of stores rises in proportion, and revenue per store does not change. If instead the additional customers raise density around existing locations, S rises through ρ: customers per store rise with $\sqrt{\rho}$, the price falls, revenue per store rises, and the number of stores rises less than in proportion to demand. Cross-border demand plausibly combines both channels, so the benchmark predicts that most, but not necessarily all, of the additional revenue appears in the number of stores; larger markets can also support larger stores for reasons outside the model (Campbell & Hopenhayn, 2005). Münter (2023) obtains a related distinction between horizontal and vertical market growth in a Cournot model with heterogeneous potential entrants. If regulation limits a market to $\bar{n} < n^*$ stores, the existing outlets serve all additional customers, customers per store rise to $S/\bar{n}$, and the adjustment moves from the store margin to revenue per store.

A convenient empirical summary of the store margin is the elasticity of market revenue with respect to the number of stores, σ. If $\sigma = 1$, an additional store adds its own sales to the market without taking any from rivals; if $\sigma = 0$, it only redistributes a fixed total. Revenue per store then moves with the number of stores at elasticity $\sigma - 1$, which is the rate at which incumbents' sales fall when a rival opens. Revenue data do not separate prices from quantities, so σ describes sales, not consumer welfare or profits (S. T. Berry & Waldfogel, 1999; Mankiw & Whinston, 1986).

3.2 Inherited capacity and the timing of entry

Free entry describes where the market is going, not how fast it gets there. Opening a dispensary requires a licensed site, a compliant build-out, supply contracts, and inspections. Firms differ in how much of this work they have already done when the market opens. Let each potential entrant choose a level of preparedness $p \in [0, 1]$, the share of its entry project completed before sales begin, at cost $K(p; A)$, where $A \geq 0$ is capacity that can be reused from earlier operations: licensed premises, compliance routines, supplier relationships, and familiarity with the regulator (Helfat & Lieberman, 2002; Prescott & Visscher, 1980). Preparedness is valuable because it brings forward the date at which a store can sell, summarized by a value $R(p)$ with $R_p > 0$ and $R_{pp} \leq 0$. The firm solves $\max_{p\in[0,1]} R(p) - K(p; A)$, with $K_p > 0$, $K_{pp} > 0$, and $K_{pA} < 0$: preparation is costly, increasingly so, and cheaper at the margin for firms with more reusable capacity.

Proposition 1 (Inherited capacity and preparedness)**.** *The problem has a unique solution $p^*(A)$, weakly increasing in A. If $p^*(A)$ is interior, it is strictly increasing, with $dp^*/dA = K_{pA}/(R_{pp} - K_{pp}) > 0$. Hence $p^*(A_H) \geq p^*(A_L)$ for $A_H > A_L$, with strict inequality whenever $p^*(A_L) < 1$ and $p^*(A_H) > 0$.*

After the market opens, unfinished projects pass through a common pipeline of licensing, inspection, and build-out. Let $\beta_k(t)$ be the additional stores of type k in an exposed market t quarters after opening, β^∞ the common long-run level, and $\phi_k = \phi(p_k^*) \in [0, 1]$, with $\phi' > 0$, the share in place at opening. If the remaining projects are completed at a common rate λ, then $\beta_k(t) = \beta^\infty[1 - (1 - \phi_k)e^{-\lambda t}]$.

Proposition 2 (Common-pipeline convergence)**.** *The difference between high- and low-capacity firms is $\beta_H(t) - \beta_L(t) = \beta^\infty(\phi_H - \phi_L)e^{-\lambda t}$. It is non-negative, declines monotonically to zero, and*

is strictly positive and strictly decreasing if $p_H^* > p_L^*$ *and* $\beta^\infty > 0$.

The proofs are in Appendix A. The two propositions separate what inherited capacity changes from what it does not. Capacity determines how much of the eventual response is in place on the first day; the common pipeline determines how fast the rest arrives; demand determines the eventual level. Two features of the setting matter for interpretation. First, A describes reusable capacity, not organizational form: a single medical dispensary has licensed premises, and a new multi-store operator need not. Whether chains have higher A is an empirical question. Second, the framework does not separate the cost of preparing from other reasons why incumbents might move first. Medical licensees may have had better access to financing, may have used their head start to occupy the best locations before rivals arrived (Fudenberg & Tirole, 1985), or may have been processed faster by a regulator that already knew them. Firms may also wait strategically when entry is irreversible and demand uncertain (Dixit, 1989). The data do not record license application or approval dates, so time spent preparing, waiting, or in regulatory review cannot be separated.

3.3 Predictions

The framework yields six predictions, which the following sections take to the data.

P1 (market expansion). Markets with more captive cross-border demand have more stores and more revenue per resident, and both rise with the amount of demand within reach.

P2 (margins of adjustment). Under free entry, the number of stores absorbs most of the additional revenue; revenue per store is unchanged if demand extends a market's reach and rises modestly if it raises density.

P3 (inherited capacity and timing). Firms with reusable capacity from the medical market open first; the gap between them and other firms shrinks as the pipeline clears.

P4 (revenue of continuing stores). Stores already open do not earn a lasting premium in high-exposure markets; their sales fall as rivals enter, at a rate governed by $\sigma - 1$.

P5 (organizational composition). An early lead of chains reflects inherited capacity; once inherited stores are separated from new entrants, chains and independents respond in proportion to demand.

P6 (entry constraints). Where the number of stores is capped, captive demand produces smaller increases in store density and larger increases in revenue per store.

P1, P2, and P4 are tested in Section 5; P3 and P5 in Section 6; the estimated model in Section 7 ties them together, and P6 is examined through a counterfactual and a descriptive comparison with Colorado in Section 8.

4 Empirical Strategy

The analysis compares nonlegal-border and always-legal-border ZIP codes quarter by quarter:

$$y_{zq} = \sum_{s=1}^{17} \beta_s \, \text{Nonlegal}_z \times \mathbb{1}[q = s] + \gamma \log(\text{Population}_z) + \delta_q + \varepsilon_{zq}, \tag{2}$$

where y_{zq} is an outcome for ZIP code z in quarter q, Nonlegal_z indicates a nonlegal-border ZIP code, and δ_q are calendar-quarter fixed effects. No quarter is omitted: each β_q is the conditional difference between the two groups in quarter q, not a change relative to 2022Q2. Imposing $\beta_q = \beta$ gives the pooled specification. Population enters because larger ZIP codes host more stores and revenue; since the outcomes are already per resident, γ captures any remaining relationship between market size and per-capita outcomes. For organizational composition, equation (2) is estimated separately for the asinh counts of independent and chain operators, N^{τ}_{zq} with $\tau \in \{I, C\}$, with type-specific quarter effects; the difference $\beta_{C,q} - \beta_{I,q}$ and its standard error come from the same regressors applied to $\text{asinh}(N^{C}_{zq}) - \text{asinh}(N^{I}_{zq})$.

The coefficients are identified from differences across ZIP codes within a quarter. Time variation shows how the gaps evolve as the market develops; it does not provide identification, because sales are observed only after the market opened and both groups gained legal sales on the same date. Let $y_{zq}(0)$ be the outcome a ZIP code would have had without captive cross-border demand. The coefficients measure the average effect of captive demand on nonlegal-border ZIP codes under two assumptions.

Assumption 1 (Conditional comparability)**.** $E[y_{zq}(0) \mid \mathit{Nonlegal}_z, \log(\mathit{Population}_z), q] = E[y_{zq}(0) \mid \log(\mathit{Population}_z), q]$.

Assumption 2 (No interference)**.** *Outcomes in always-legal-border ZIP codes do not depend on the captive demand available to nonlegal-border ZIP codes.*

Assumption 1 is the substantive restriction. Both groups lie near a state line, which removes generic border effects, but they occupy different parts of the state. Nonlegal-border ZIP codes are concentrated in the southeast; always-legal-border ZIP codes are in the northwest, where many communities lie on tribal lands that regulate cannabis under their own authority. The groups may also differ in remoteness, in the size of their medical markets, and in general commercial activity. Assumption 2 would fail if Texas residents drove past nonlegal-border markets to buy in always-legal-border ZIP codes; given distances of more than 75 miles, this is unlikely to matter. Because Assumption 1 cannot be verified directly, the estimates are described as conditional spatial differences, and the assumption is assessed in Section 5.5 with predetermined controls, an imputation estimator, exposure intensity, and placebo sectors.

Standard errors are clustered by county. Because the main sample has 23 counties, the paper also reports wild cluster restricted bootstrap p-values with six-point weights (Cameron, Gelbach, & Miller, 2008; Webb, 2023), Conley standard errors that allow spatial correlation between ZIP codes up to 50 or 100 miles apart (Conley, 1999), and estimates clustered by town market. Bootstrap procedures use at least 999 draws and fixed seeds, which are recorded in the replication package.

5 Cross-Border Demand and Market Adjustment

5.1 Revenue, store density, and revenue per store

Markets that can draw on captive cross-border demand are larger in every quarter. Table 2 and Figure 2 report the quarterly coefficients of equation (2). The gap in asinh revenue per resident is positive and significant in all 17 quarters; the gap in asinh dispensary density is positive throughout and significant from 2022Q4; and the gap in log revenue per store is positive in most quarters but never significant. Because these are asinh and log coefficients on outcomes with many zeros and small units, magnitudes are read from other scales. In Poisson regressions with the same controls, nonlegal-border ZIP codes generate 2.44 times the revenue of always-legal-border ZIP codes (coefficient 0.892, SE 0.281) and host 1.94 times as many dispensaries (0.664, SE 0.240). Estimated quarter by quarter, the revenue ratio ranges from 2.18 in 2022Q3 to 2.59 in 2026Q2,

while the store ratio rises from 1.62 to a peak of 2.16 in 2024Q3 and falls to 1.63 by 2026Q2 (Online Resource 1, Table OR3). In levels, pooling all quarters, nonlegal-border ZIP codes have $45 more quarterly revenue per resident (SE 17), against a mean of $12 in always-legal-border ZIP codes, and are 15 percentage points more likely to host any dispensary (SE 5), against a mean of 13 percent. These patterns are consistent with P1.

Inference does not hinge on the small number of clusters. The wild cluster bootstrap p-values for the pooled gaps are 0.004 for revenue per resident, 0.035 for density, and 0.238 for revenue per store; Conley standard errors and standard errors clustered by town market are similar to the county-clustered ones, and all 23 leave-one-county-out estimates of the revenue gap are significant at the 5 percent level (Table OR4).

Table 2: Nonlegal- versus always-legal-border gaps in market outcomes, by quarter

Quarter	Revenue per resident (asinh)		Dispensary density (asinh)		log(Revenue per store)	
	Coef.	(SE)	Coef.	(SE)	Coef.	(SE)
2022Q2	+0.459**	(0.203)	+0.017	(0.034)	-0.355	(0.753)
2022Q3	+0.583**	(0.240)	+0.039	(0.029)	-0.322	(0.238)
2022Q4	+0.729***	(0.218)	+0.063*	(0.030)	+0.094	(0.331)
2023Q1	+0.832***	(0.252)	+0.084**	(0.035)	+0.298	(0.370)
2023Q2	+0.868***	(0.257)	+0.105***	(0.036)	+0.231	(0.287)
2023Q3	+0.904***	(0.280)	+0.117**	(0.048)	+0.415	(0.414)
2023Q4	+0.944***	(0.292)	+0.135**	(0.049)	+0.348	(0.328)
2024Q1	+0.951***	(0.295)	+0.152**	(0.058)	+0.286	(0.347)
2024Q2	+0.954***	(0.304)	+0.166***	(0.058)	+0.337	(0.370)
2024Q3	+0.961***	(0.306)	+0.171***	(0.058)	+0.332	(0.506)
2024Q4	+0.960***	(0.293)	+0.174***	(0.055)	+0.229	(0.192)
2025Q1	+0.900***	(0.308)	+0.145**	(0.070)	+0.519	(0.341)
2025Q2	+0.818**	(0.303)	+0.114	(0.068)	+0.336	(0.214)
2025Q3	+0.818**	(0.312)	+0.114*	(0.063)	+0.027	(0.239)
2025Q4	+0.846**	(0.304)	+0.097	(0.061)	+0.129	(0.234)
2026Q1	+0.817**	(0.309)	+0.085	(0.062)	+0.186	(0.229)
2026Q2	+0.858***	(0.299)	+0.081	(0.062)	+0.664	(0.442)

Notes: Each cell reports $\hat{\beta}_q$ from equation (2): the conditional gap between nonlegal-border and always-legal-border ZIP codes in quarter q, not a change relative to a reference quarter. Sample: 92 nonlegal-border and 117 always-legal-border ZIP codes, 2022Q2–2026Q2 (3,553 ZIP-quarters). Revenue per resident (quarterly medical plus adult-use sales divided by population) and dispensary density (dispensaries per 1,000 residents) are in inverse-hyperbolic-sine units and include ZIP-quarters with no dispensary; the coefficients are not percentage effects. Revenue per store is in logs and uses the 755 ZIP-quarters with at least one dispensary. All regressions include log population and calendar-quarter fixed effects. Standard errors clustered by county (23 clusters). $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

5.2 How much of the revenue gap is store count?

Because revenue equals the number of stores times revenue per store, the gap in log revenue per resident between the two groups splits exactly into a store-density component and a revenue-per-

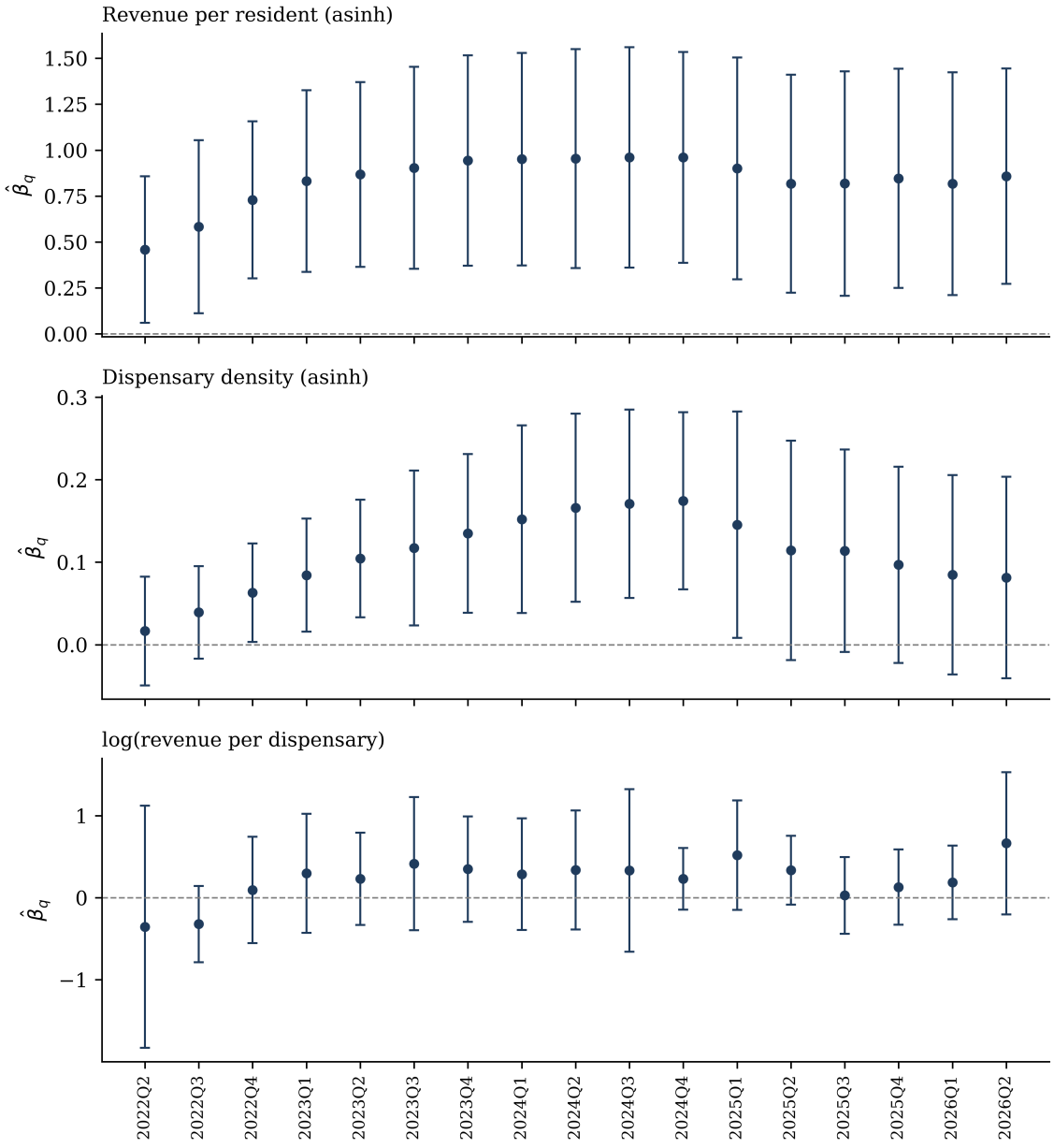


Fig. 2 Quarterly gaps in revenue per resident, dispensary density, and revenue per dispensary

Notes: Point estimates and 95% confidence intervals for $\hat{\beta}_q$ in Table 2. Same sample, specification, and county-clustered standard errors.

store component:

$$\underbrace{\log \frac{R_{NL}}{P_{NL}} - \log \frac{R_{LB}}{P_{LB}}}_{\text{total}} = \underbrace{\log \frac{N_{NL}}{P_{NL}} - \log \frac{N_{LB}}{P_{LB}}}_{\text{store density}} + \underbrace{\log \frac{R_{NL}}{N_{NL}} - \log \frac{R_{LB}}{N_{LB}}}_{\text{revenue per store}}, \tag{3}$$

where R, N, and P are total revenue, operating stores, and population in each group and quarter. Group totals include ZIP-quarters without a store, which contribute population but no stores or revenue, so zeros require no transformation. The decomposition is an accounting identity: it describes how the observed gap divides, not what would happen if the number of stores were changed. Intervals come from 999 county-cluster bootstrap draws that resample counties within each group.

Store density accounts for most of the gap (Table 3 and Figure 3). Pooling all quarters, the total log gap is 1.04, of which 0.76 is store density and 0.29 is revenue per store; the store share is 73 percent, with a 95 percent interval from 38 to 118 percent. The store share lies between 62 and 86 percent in every quarter through 2025Q3. It falls to 53 percent in 2026Q2, when the revenue-per-store component reaches 0.52, the only quarter in which that component is significant. The revenue-per-store margin is therefore imprecisely estimated rather than shown to be zero: the pooled regression gap of 0.26 log points has a 95 percent interval from -0.16 to 0.68. What the data do reject is the benchmark in which existing stores carry the whole gap, as they would if entry were fixed: the pooled revenue-per-store gap is smaller than the total gap of 1.04 (one-sided $p = 0.001$), and no bootstrap draw attributes the whole gap to revenue per store. This is the pattern P2 predicts when cross-border demand mainly extends a market's reach.

5.3 Adult-use or medical sales?

If the gap reflects captive cross-border demand, it should appear in adult-use sales, which any adult can buy, rather than in medical sales, which require a New Mexico medical card. Table 4 splits revenue accordingly. Medical sales are 18 percent of revenue in both groups of ZIP codes. The adult-use revenue ratio from the Poisson regression is 2.66 (coefficient 0.980, SE 0.298), close to the total, whereas the medical ratio is 1.45 and not significant (0.375, SE 0.354). Store density accounts for 72 percent of the adult-use gap, almost exactly the share for total revenue. The medical

Table 3: Exact decomposition of the gap in revenue per resident

Period	$\Delta \log(R/P)$ (total)	$\Delta \log(N/P)$ (stores)	$\Delta \log(R/N)$ (revenue per store)	Share via store count
2022Q2	+0.937	+0.715	+0.222	0.76
	[+0.28, +2.01]	[+0.13, +1.55]	[-0.59, +1.02]	[0.19, 2.42]
2023Q2	+1.053	+0.854	+0.199	0.81
	[+0.54, +1.73]	[+0.37, +1.38]	[-0.27, +0.64]	[0.50, 1.36]
2024Q2	+1.085	+0.805	+0.280	0.74
	[+0.52, +1.74]	[+0.26, +1.32]	[-0.16, +0.78]	[0.34, 1.19]
2025Q2	+1.057	+0.790	+0.267	0.75
	[+0.50, +1.69]	[+0.26, +1.29]	[-0.16, +0.71]	[0.36, 1.22]
2026Q2	+1.103	+0.583	+0.520	0.53
	[+0.50, +1.72]	[+0.10, +1.00]	[+0.08, +0.94]	[0.12, 0.89]
Pooled	+1.043	+0.757	+0.286	0.73
	[+0.48, +1.71]	[+0.29, +1.22]	[-0.12, +0.76]	[0.38, 1.18]

Notes: Each row decomposes the difference between nonlegal- and always-legal-border ZIP codes in log revenue per resident, computed from group totals of revenue (R), operating stores (N), and population (P), into store density and revenue per store, as in equation (3). The last column is the store-density component divided by the total. The pooled row sums revenue, stores, and population over all 17 quarters. Brackets report 95% intervals from 999 county cluster bootstrap draws, resampling counties within each group. Sample: 209 ZIP codes, including ZIP-quarters without stores.

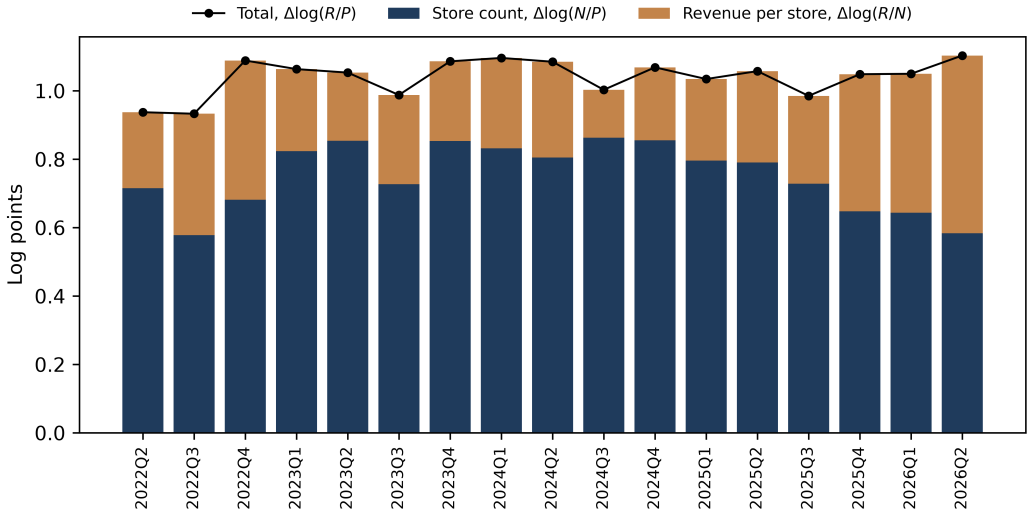


Fig. 3 Store-density and revenue-per-store components of the revenue gap, by quarter

Notes: Bars show the two components of equation (3); the line shows their sum. Values from Table 3.

revenue-per-resident gap in asinh units is positive (0.541, SE 0.160), as expected if more stores also make medical purchases more convenient, but it is the adult-use market that carries the border premium. The gap does not depend on stores inherited from the medical market either: counting only storefronts that were not medical incumbents, the pooled gaps are 0.821 (SE 0.237) in asinh revenue per resident and 0.105 (SE 0.042) in density, essentially the baseline values.

Table 4: Total, adult-use, and medical revenue

	Total	Adult-use	Medical
Panel A: Regression gaps (209 ZIP codes)			
Revenue per resident (asinh)	$+0.835^{***}$	$+0.799^{***}$	$+0.541^{***}$
	(0.263)	(0.254)	(0.160)
Revenue per dispensary (log)	+0.260	+0.260	+0.130
	(0.199)	(0.236)	(0.232)
Revenue, Poisson coefficient	$+0.892^{***}$	$+0.980^{***}$	+0.375
	(0.281)	(0.298)	(0.354)
Implied revenue ratio	2.44	2.66	1.45
Panel B: Exact decomposition of the log gap in revenue per resident, pooled			
Total gap	1.043	1.052	1.004
	[0.60, 1.64]	[0.54, 1.71]	[0.50, 1.83]
Store-density component	0.757	0.757	0.757
Revenue-per-store component	0.286	0.295	0.248
	[-0.08, 0.74]	[-0.14, 0.77]	[-0.02, 0.78]
Store-density share	0.73	0.72	0.75
	[0.50, 1.11]	[0.44, 1.19]	[0.53, 1.04]
Share of revenue	1.00	0.82	0.18

Notes: Panel A: pooled versions of equation (2) for each revenue stream, with log population and quarter fixed effects; revenue per dispensary uses ZIP-quarters with at least one dispensary and positive revenue in the stream; the Poisson rows use ZIP-quarter revenue. County-clustered standard errors in parentheses (23 clusters). Panel B: exact decomposition of equation (3) for each stream, pooling all quarters; the store-density component is the same for every stream because it counts all operating dispensaries. Brackets report 95% intervals from 999 county cluster bootstrap draws (seed 20261023). Medical sales are those reported as medical by the licensee. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

5.4 Continuing stores and the arrival of rivals

ZIP-level revenue per store averages over continuing stores and new entrants. The dispensary panel separates them, which matters for interpretation: if continuing stores earned a premium in exposed markets that eroded as rivals arrived, the market adjustment would involve temporary rents to incumbents; if they never earned one, the adjustment ran through entry from the start. Table 5 and Figure 4 examine the 304 storefronts selling in 2022Q2.

There is no initial premium. In 2022Q2, log revenue of continuing stores is unrelated to the log number of nonlegal-state residents within 75 miles (0.044, SE 0.032), and continuing stores in

nonlegal-border ZIP codes earned no more than those in always-legal-border ZIP codes (−0.06, SE 0.43). With store and quarter fixed effects, the revenue of continuing stores in high-exposure markets did not rise relative to opening; if anything it drifted down late in the sample (late-period slope −0.051, SE 0.043). What does predict lower sales is the arrival of rivals: within store, a one-log-point increase in one plus the number of other dispensaries within 2 kilometers is associated with revenue lower by 0.24 log points (SE 0.09), and the association is stronger in higher-exposure markets. Rival entry is itself a response to local conditions, so this is a conditional association rather than a causal estimate of business stealing, but its sign is the opposite of what entry into growing markets alone would produce, and it agrees with evidence from grocery retail that entry lowers incumbents' margins even when their sales hold up (Chenarides, Gómez, Richards, & Yonezawa, 2024). Survival of continuing stores to 2026Q2 does not vary with exposure.

The same picture holds for all stores. Comparing the 447 dispensaries located in nonlegal-border areas with the 80 in always-legal-border areas, with exposure defined at the store's own location, log revenue differs by −0.11 (SE 0.26) and log revenue per resident within 1 kilometer by −0.10 (SE 0.22), and the estimates remain negative and insignificant with operator fixed effects (Table OR11 and Figure OR1). Stores near a nonlegal-state line do not earn detectably more. Together with the decomposition, this evidence supports P4: captive demand was served by more stores, not by lasting higher sales at the stores already in place.

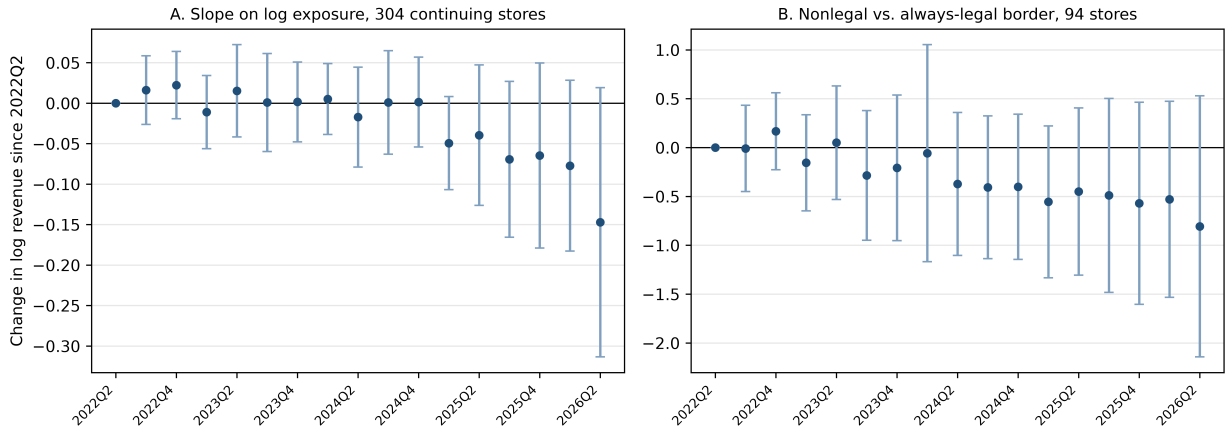


Fig. 4 Revenue of continuing stores relative to market opening

Notes: Coefficients and 95% confidence intervals from regressions of log quarterly revenue of the stores selling in 2022Q2 on store fixed effects, quarter fixed effects, and interactions of quarter with (a) log nonlegal-state residents within 75 miles or (b) a nonlegal-border indicator; 2022Q2 is the omitted quarter. County-clustered standard errors.

Table 5: Continuing stores: initial revenue, revenue path, and local rivals

Outcome and regressor	Estimate	Observations
Log revenue in 2022Q2: log exposure (statewide)	+0.044	304
	(0.032)	
Log revenue in 2022Q2: nonlegal border (main sample)	−0.062	94
	(0.434)	
Change since 2022Q2, late mean: log exposure	−0.051	304
	(0.043)	
Change since 2022Q2, late mean: nonlegal border	−0.509	94
	(0.445)	
Change since 2022Q2, late mean: log exposure, adult-use revenue	−0.050	304
	(0.040)	
Log(1 + rivals within 2 km), store and quarter FE	-0.236^{***}	3,885
	(0.086)	
Log(1 + rivals within 5 km), store and quarter FE	-0.323^{***}	3,885
	(0.108)	
× log exposure	−0.017	3,885
	(0.016)	
Survived to 2026Q2 (linear probability): log exposure	+0.011	304
	(0.009)	

Notes: Continuing stores are the 304 storefronts reporting sales in 2022Q2 (94 in the 209 main-sample ZIP codes: 78 nonlegal-border, 16 always-legal-border). Rows 1–2: cross-section of 2022Q2 log revenue on log exposure, $\log(1 + E_z/1000)$, or a nonlegal-border indicator, controlling for log residents within 1 km. Rows 3–5: average over 2024Q2–2026Q2 of the coefficients on exposure (or the indicator) interacted with quarter, in regressions of log revenue on store fixed effects, quarter fixed effects, and these interactions (2022Q2 omitted). Rows 6–8: log revenue on log(1 + other active dispensaries within r km) with store and quarter fixed effects. Row 9: linear probability model for operating in 2026Q2. Standard errors clustered by county in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

5.5 Threats to identification

Both groups lie near a state line, but they are not identical places. Table 6 compares predetermined characteristics. Nonlegal-border ZIP codes are larger (5,700 versus 1,900 residents), farther from Albuquerque (194 versus 128 miles), and more likely to have hosted a medical dispensary in April 2022 (17 versus 4 percent). They do not differ significantly in population density, in distance to the nearest New Mexico city of at least 50,000 residents, in retail establishments per resident in 2019, or in medical incumbents per resident. If the cannabis gap reflected these differences rather than captive demand, four things should be true: the gap should disappear once the differences are held fixed; places that never faced captive demand should be able to predict it; it should be unrelated to how much captive demand a ZIP code faces; and it should appear in other kinds of retail. Each is examined below.

Table 6: Predetermined characteristics of nonlegal- and always-legal-border ZIP codes

	Nonlegal-border	Always-legal-border	Difference (SE)	Normalized difference
Resident population (000s)	5.69	1.88	$+3.81^{***}$ (1.01)	+0.48
log population density	2.13	1.44	+0.69 (0.56)	+0.42
Distance to Albuquerque (mi)	193.91	128.42	$+65.49^{***}$ (14.73)	+1.79
Distance to nearest NM city ≥50k (mi)	61.59	78.27	−16.68 (16.06)	-0.44
Retail establishments per 1,000, 2019 (ZBP)	1.70	2.37	−0.67 (1.23)	-0.08
Medical incumbents per 1,000, April 2022	0.03	0.02	+0.01 (0.02)	+0.10
Any medical incumbent, April 2022	0.17	0.04	$+0.13^{***}$ (0.03)	+0.43
Nonlegal-state residents within 75 mi (000s)	288.17	0.02	$+288.15^{**}$ (132.10)	+1.24
Legal-state residents within 75 mi (000s)	11.04	30.80	-19.75^{**} (8.94)	-0.73
ZIP codes	92	117		

Notes: ZIP-code level. Columns 1–2 report means. Column 3 reports the difference from a regression on a nonlegal-border indicator with county-clustered standard errors in parentheses. Column 4 reports the difference in means divided by the square root of the average of the two group variances. Population density from SimpleMaps; distances computed from ZIP-code centroids, with city centroids defined as population-weighted means of their ZIP codes; retail establishments from the 2019 ZIP Codes Business Patterns; medical incumbents are storefronts reporting medical sales in April 2022. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Predetermined controls. Panel A of Table 7 adds the characteristics to the pooled regression. Holding fixed population density, distance to Albuquerque, distance to the nearest large city, and 2019 retail activity, the revenue-per-resident gap falls from 0.84 to 0.48 and the density gap from 0.11 to 0.08; the revenue gap remains significant at the 10 percent level. Density, distance to the nearest large city, 2019 retail activity, and medical incumbents, each added alone, leave the revenue

gap between 0.75 and 0.94 and significant at the 1 percent level (Table OR5). Almost all of the decline comes from distance to Albuquerque, which in this sample largely measures closeness to Texas: every nonlegal-border ZIP code lies more than 140 miles from Albuquerque, against only 31 of the 117 always-legal-border ZIP codes. Among the 276 ZIP codes with no nonlegal-state exposure, where distance to Albuquerque varies for other reasons, it predicts none of the outcomes (0.07, SE 0.18, for revenue per resident). A control for longitude behaves the same way: its correlation with the nonlegal indicator is 0.46, and adding it lowers the revenue gap to 0.40 (SE 0.20) and the density gap to 0.04 (SE 0.04) (Table OR7). Medical incumbents are a weak control for a different reason: their stores are part of the 2022Q2 outcome, and licensees could anticipate legalization once the Act passed, so holding them fixed partly holds fixed the response itself. With all controls the revenue gap is 0.22 (SE 0.25).

Imputation from unexposed ZIP codes. Regression controls fit the effect of distance to Albuquerque mainly from the treated group, where it moves almost one for one with exposure. An imputation estimator avoids this (in the spirit of Borusyak, Jaravel, & Spiess, 2024). It learns how outcomes depend on the controls using only the 276 always-legal-border and interior ZIP codes, predicts what each nonlegal-border ZIP code would have earned under that relationship, and averages the difference between actual and predicted outcomes. Panel B of Table 7 reports the results. With population alone, the gaps are 0.58 (bootstrap SE 0.25) for revenue per resident and 0.095 (0.043) for density; with geographic and retail controls they are 0.48 (0.28) and 0.090 (0.050); and restricting to nonlegal-border ZIP codes whose distance to Albuquerque is also observed among always-legal-border ZIP codes gives 0.42 and 0.081. Measured against places that never faced captive demand, nonlegal-border ZIP codes still sell more cannabis and host more dispensaries per resident, by about one-half to two-thirds of the baseline gaps, significant at about the 10 percent level.

Exposure intensity. If captive demand drives the gap, ZIP codes facing more of it should respond more. Exposure varies widely within the nonlegal-border group, from about 2,700 to 800,000 nonlegal-state residents within 75 miles. Table 8 replaces the binary indicator with $\log(1 + E_z/1000)$. Comparing nonlegal-border ZIP codes with one another and holding geography and

Table 7: The pooled nonlegal-border gap with predetermined controls and imputation

	Revenue per resident (asinh) (1)	Dispensaries per 1,000 (asinh) (2)	log revenue per dispensary (3)
Panel A: Regression with predetermined controls (209 ZIP codes, 92 nonlegal border)			
Baseline (pooled Table 3 specification)	+0.835***	+0.109**	+0.260
	(0.263)	(0.045)	(0.199)
+ geography	+0.448	+0.075	+0.348
	(0.347)	(0.072)	(0.287)
+ geography + 2019 retail	+0.475*	+0.081	+0.320
	(0.263)	(0.052)	(0.262)
+ geography + 2019 retail + medical incumbents	+0.224	+0.033	+0.171
	(0.249)	(0.051)	(0.213)
Panel B: Imputation from the 276 unexposed ZIP codes			
Log population only	+0.576**	+0.095**	+0.293*
	(0.245)	(0.043)	(0.169)
+ geography	+0.455	+0.086	+0.260
	(0.311)	(0.054)	(0.396)
+ geography + 2019 retail	+0.477	+0.090*	+0.335
	(0.283)	(0.050)	(0.380)
+ geography + 2019 retail, common support	+0.418	+0.081*	+0.269
	(0.270)	(0.049)	(0.382)

Notes: Panel A: pooled version of equation (2) with log population and calendar-quarter fixed effects; county-clustered standard errors (23 clusters). Panel B: the outcome is regressed on the controls, log population, and quarter fixed effects in the 276 always-legal-border and interior ZIP codes; the estimate is the average, over nonlegal-border ZIP-quarters, of the observed minus the predicted outcome; standard errors from 499 county-cluster bootstrap draws. Common support keeps the 90 nonlegal-border ZIP codes whose distance to Albuquerque lies within the range observed among always-legal-border ZIP codes. Geography: log population density, distance to Albuquerque, and distance to the nearest New Mexico city with at least 50,000 residents. 2019 retail: asinh of retail establishments per 1,000 residents. Medical incumbents: asinh of storefronts with medical sales in April 2022 per 1,000 residents. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

retail activity fixed, a one-log-point increase in exposure raises asinh revenue per resident by 0.18 (SE 0.04) and asinh density by 0.042 (SE 0.006). Without the controls the within-group slope is imprecise (0.12, SE 0.13), so this comparison relies on holding geography fixed. Across all 368 ZIP codes the slope is significant with or without controls. Measuring reach by road-network travel time gives the same pattern (Table OR8): the density gradient is positive and significant for 60-, 90-, and 120-minute catchments and for a travel-time decay measure, in all three samples. In none of these specifications does revenue per store respond to exposure at the 5 percent level. Splitting exposure by state shows that the gradient comes from Texas and Oklahoma; residents of Utah within 75 miles, who are concentrated in the sparsely populated and largely tribal Four Corners region, are associated with lower outcomes (Table OR7).

Table 8: Exposure intensity and cannabis outcomes

	Revenue per resident (asinh) (1)	Dispensaries per 1,000 (asinh) (2)	log revenue per dispensary (3)
Panel A: Within the 92 nonlegal-border ZIP codes (13 county clusters)			
No controls	+0.117	+0.021	+0.019
	(0.130)	(0.021)	(0.028)
+ geography + 2019 retail	+0.178***	+0.042***	−0.001
	(0.041)	(0.006)	(0.057)
Panel B: All 368 New Mexico ZIP codes (33 county clusters)			
No controls	+0.125**	+0.021***	+0.052*
	(0.047)	(0.007)	(0.029)
+ geography + 2019 retail	+0.142***	+0.026***	+0.048
	(0.045)	(0.009)	(0.042)
+ geography + 2019 retail + medical incumbents	+0.121***	+0.022***	+0.048
	(0.038)	(0.008)	(0.040)

Notes: Each cell is the coefficient on $\log(1 + E_z/1000)$, where E_z is the number of residents of Texas, Oklahoma, and Utah within 75 miles of the ZIP-code centroid, from the pooled version of equation (2) with the binary indicator replaced by this measure. Log population and quarter fixed effects in all columns; controls as in Table 7. County-clustered standard errors in parentheses; p-values use $G - 1$ degrees of freedom. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Other definitions, tribal lands, and influential markets. The binary gap is robust to the population threshold in the exposure rule (0.81 to 1.00 for thresholds from 500 to 10,000 residents), to dropping the 14 ZIP codes exposed to both kinds of border (0.97), and to dropping or controlling for the 39 ZIP codes whose centroid lies on tribal land (0.83 and 0.71; U.S. Census Bureau, 2024) (Tables OR7 and OR8). It is larger with a 50-mile radius (1.13) and smaller with a 100-mile radius (0.66), as expected if exposure declines with distance, and it remains significant when interior

ZIP codes join the comparison group. Dropping either group's largest county, the Colorado- or Arizona-exposed comparison ZIP codes, or the 5 percent of ZIP codes with the highest revenue per resident leaves revenue gaps between 0.54 and 1.01, all significant. Dropping the 80 nonlegal-border ZIP codes closest to Texas eliminates the gap; the remaining 12 face a median of about 3,700 nonlegal-state residents within 75 miles, against 132,000 for the Texas-exposed group, so the binary comparison is identified by the Texas border (Table OR4).

Placebo sectors. The cannabis data begin when sales began, so they cannot show what these areas looked like before legalization. Other retail sectors can. Table 9 estimates the same comparison for grocery stores and total retail establishments per resident, using ZIP Codes Business Patterns data for 2018–2023. Nonlegal-border ZIP codes do not have significantly more grocery or retail establishments per resident in any year, and the gap does not change significantly after legalization (0.015, SE 0.010, for grocery; 0.045, SE 0.061, for total retail). One pattern calls for caution: the total-retail gap rises from 0.19 in 2018 to 0.46 in 2021, with no single year significant, which suggests that nonlegal-border areas were on a somewhat different commercial path before legalization. That rise stops after 2021, when the cannabis gaps open. Alcohol retail establishments are too rare in these ZIP codes to be informative.

Taken together, the evidence supports P1, P2, and P4 with a qualification about magnitude. Markets that can draw on residents of a prohibition state host more dispensaries and generate more cannabis revenue per resident; the difference is concentrated in adult-use sales and rises with the accessible out-of-state population; and most of it is accounted for by additional stores, while continuing stores earned no premium. Controls that capture geography and the imputation estimator reduce the gap to roughly half of the baseline, so the baseline is best treated as the upper end of the plausible range.

Table 9: Placebo: non-cannabis retail establishments, 2018–2023

	Alcohol retail	Grocery stores	Total retail
Panel A: Gap in asinh(establishments per 10,000 residents) by year			
2018	−0.003	+0.001	+0.185
	(0.017)	(0.052)	(0.271)
2019	−0.015	+0.015	+0.322
	(0.012)	(0.055)	(0.289)
2020	−0.015	+0.020	+0.384
	(0.012)	(0.055)	(0.305)
2021	−0.015	+0.020	+0.456
	(0.012)	(0.054)	(0.307)
2022 (post)	−0.013	+0.023	+0.363
	(0.011)	(0.054)	(0.292)
2023 (post)	−0.013	+0.034	+0.400
	(0.011)	(0.052)	(0.297)
Panel B: Change in the gap after legalization			
Nonlegal × Post (2022–23 vs. 2018–21)	−0.001	+0.015	+0.045
	(0.003)	(0.010)	(0.061)
Always-legal-border mean of outcome	0.012	0.068	1.257
ZIP codes / years		209 / 6	

Notes: ZIP Codes Business Patterns establishment counts for alcohol retail (NAICS 445310), grocery stores (NAICS 4451), and total retail trade (NAICS 44–45), 2018–2023, for the 209 ZIP codes of the main sample; missing ZIP-year-sector cells are zeros. Outcome: asinh of establishments per 10,000 residents. Panel A: year-specific nonlegal-border gaps with log population and year fixed effects (no omitted year). Panel B: coefficient on Nonlegal × Post from a regression that also includes Nonlegal, year fixed effects, and log population. Alcohol retail has seven nonzero ZIP-years in this sample and is uninformative. Standard errors clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

6 Incumbency, Organizational Form, and Entry Timing

6.1 Who supplied the first stores?

Table 10 and Figure 5 estimate equation (2) separately for chain and independent operators under the lifetime classification. In 2022Q2, the first quarter of adult-use sales, the nonlegal-border gap is 0.22 (SE 0.08) for chain operators and 0.06 (SE 0.08) for independents. In operator counts, the fitted model implies 0.43 chain operators in the average nonlegal-border ZIP code, against 0.16 had it responded like an always-legal-border ZIP code of the same population, and 0.15 independent operators against 0.08. Per 100 nonlegal-border ZIP codes, captive demand was met at opening by about 26 additional chain operators (95 percent county-bootstrap interval 8 to 45) and 6 additional independents, an estimate not distinguishable from zero (−12 to 24); the chain advantage of 20 operators per 100 ZIP codes is precisely estimated (11 to 30). The chain lead averages 0.11 (SE 0.03) over the first year.

The first suppliers were the stores already in place. In 2022Q2, 19 nonlegal-border ZIP codes had a dispensary, against 7 always-legal-border ZIP codes, and most of these stores were medical dispensaries of multi-store operators that converted to adult-use sales on the first day. Among ZIP-quarters with at least one dispensary, with log outcomes and market-age fixed effects, the 2022Q2 chain gap is 0.10 (SE 0.20) and the chain-minus-independent difference 0.05 (SE 0.30) (Table OR14). Nonlegal-border ZIP codes were thus more likely to have a chain store when sales began, but markets that had stores did not have more chain operators. Because whether a ZIP code has a dispensary is itself affected by the border, this conditional comparison is descriptive.

The lead then closes. From 2023Q2 onward no quarterly difference is significant, and over the last nine quarters the chain-minus-independent difference averages 0.01, with a 90 percent confidence interval from −0.06 to 0.09; a two-one-sided test rejects that this late difference is as large in absolute value as the early lead ($p = 0.019$; Schuirmann, 1987). Panel B of Figure 5 adds ZIP-by-type fixed effects, which absorb any time-invariant difference in how attractive a ZIP code is to each type; the difference falls by 0.15 (SE 0.04) between 2022Q2 and 2024Q2–2026Q2. The closing is therefore itself significant and not driven by persistent differences across markets.

Table 10: Independent and chain operators, lifetime classification, by quarter

Quarter	N independent brands (1)		N chain brands (2)		Indep. − chain (3)	
	Coef.	(SE)	Coef.	(SE)	Coef.	(SE)
2022Q2	+0.055	(0.078)	+0.219**	(0.079)	-0.173***	(0.036)
2022Q3	+0.127*	(0.066)	+0.243***	(0.078)	-0.124***	(0.035)
2022Q4	+0.204***	(0.071)	+0.280***	(0.075)	-0.084*	(0.043)
2023Q1	+0.240***	(0.067)	+0.321***	(0.079)	-0.089**	(0.038)
2023Q2	+0.294***	(0.061)	+0.342***	(0.077)	-0.056	(0.046)
2023Q3	+0.331***	(0.058)	+0.343***	(0.085)	-0.020	(0.043)
2023Q4	+0.357***	(0.061)	+0.358***	(0.082)	-0.009	(0.037)
2024Q1	+0.371***	(0.062)	+0.367***	(0.080)	-0.004	(0.029)
2024Q2	+0.365***	(0.064)	+0.372***	(0.087)	-0.015	(0.042)
2024Q3	+0.380***	(0.066)	+0.373***	(0.085)	-0.001	(0.040)
2024Q4	+0.364***	(0.059)	+0.387***	(0.083)	-0.030	(0.044)
2025Q1	+0.371***	(0.066)	+0.375***	(0.084)	-0.012	(0.048)
2025Q2	+0.345***	(0.063)	+0.380***	(0.086)	-0.043	(0.051)
2025Q3	+0.372***	(0.062)	+0.367***	(0.085)	-0.003	(0.052)
2025Q4	+0.335***	(0.060)	+0.359***	(0.077)	-0.032	(0.038)
2026Q1	+0.334***	(0.059)	+0.343***	(0.076)	-0.017	(0.042)
2026Q2	+0.316***	(0.067)	+0.337***	(0.075)	-0.028	(0.038)

Notes: Quarter-specific nonlegal- versus always-legal-border coefficients. Outcomes: asinh of the number of distinct independent and chain operators active in the ZIP code; a chain is an operator with two or more New Mexico storefronts at any point in the sample. Columns 1–2 are separate regressions with log population and quarter fixed effects. Column 3 reports the independent-minus-chain difference from a stacked regression with type-specific quarter fixed effects and a common log-population slope (Figure 5 plots the same difference with the opposite sign). Sample: 209 ZIP codes, 3,553 ZIP-quarters per type. Standard errors clustered by county (23 clusters). $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

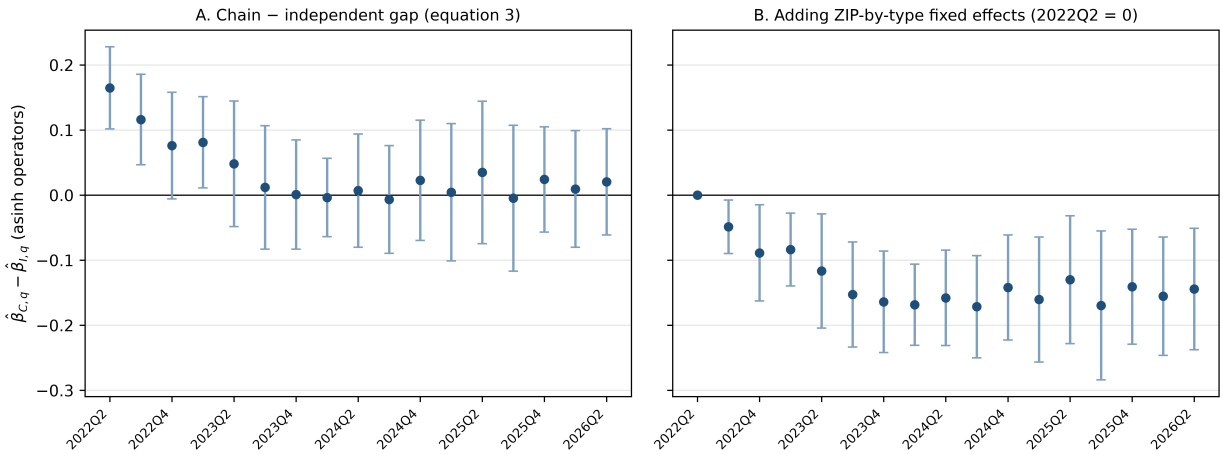


Fig. 5 Chain-minus-independent difference in the nonlegal-border response

Notes: Each point is $\hat{\beta}_{C,q} - \hat{\beta}_{I,q}$, the nonlegal-border coefficient for asinh chain operators minus that for asinh independent operators, with 95% confidence intervals from county-clustered standard errors. Chain: operator with two or more New Mexico storefronts at any point in 2022Q2–2026Q2, including medical incumbents and operators that expand after opening a single store. Independent: operator with one storefront throughout. Panel (a): type-specific quarter fixed effects and log-population slopes, no quarter omitted. Panel (b): adds ZIP-by-type fixed effects and omits 2022Q2; each coefficient is the change in the difference relative to 2022Q2. 209 ZIP codes, 3,553 ZIP-quarters per type.

6.2 Three classifications and the inherited stock

The lifetime classification has two features that matter for timing. It uses future information: an operator that opens its first store in 2022 and its second in 2025 is a chain from 2022 onward. And it treats stores that converted from the medical market in the same way as new entrants. It is therefore the classification most favorable to finding a chain lead, because it credits every operator that ever grows as a chain from its first day; if chains had expanded faster than independents after legalization, this classification would show it most clearly. The classification at market opening errs in the opposite direction, because successful newcomers stay in the independent group even after they become chains. Under the status-at-entry classification, every store operating in 2022Q2 is first observed in the same quarter and therefore counts as a new-operator entry, so this classification is informative only for stores that opened later. Table 11 and Figure 6 report all three, with and without the stores operating in 2022Q2.

The early chain lead comes from the stock in place on the first day. Of the 78 storefronts operating in nonlegal-border ZIP codes in 2022Q2, 61 belonged to lifetime chains and 45 were medical incumbents; the always-legal-border ZIP codes had 16 storefronts, 14 run by chains and 8 by medical incumbents. Statewide, 188 of the 226 medical-incumbent storefronts (83 percent) belonged to operators that already ran two or more stores at opening, and only 13 of the 38 single-store medical operators later opened another store. Once the stores operating in 2022Q2 are dropped, the nonlegal-border gap in new chain operators matches the gap in new independent operators: the difference averages -0.005 (SE 0.021) over the first year and 0.020 (SE 0.037) over 2024Q2–2026Q2. Even under the classification most generous to chains, stores opened after legalization show no chain lead.

Classified at market opening, the early lead is small and later reverses. The 2022Q2 gap is 0.20 (SE 0.05) for chains and 0.08 (SE 0.10) for independents; over the first year the difference averages 0.01 (SE 0.05), and from 2023 onward the independent gap exceeds the chain gap by 0.27 (SE 0.07), significant in all nine late quarters. Part of this late independent advantage reflects firms that started small after opening and then expanded, which this classification keeps among the independents; the same firms move into the chain group under the lifetime classification as they grow. Among stores opened after the market did, status at entry shows that new operators' first

stores responded at least as strongly as established operators' additional stores (differences of 0.09, SE 0.04, over the first year and 0.14, SE 0.05, later).

Table 11: Independent-minus-chain differences in the nonlegal-border gap, by classification

	2022Q2		Early mean	Late mean	TOST	Late quarters
Classification	Indep.	Chain	$I - C$	$I - C$	p-value	with $p < .05$
Lifetime	+0.055	+0.219**	−0.109***	−0.012	0.019	0/9
	(0.078)	(0.079)	(0.032)	(0.044)		
Predetermined at opening	+0.081	+0.203***	+0.008	+0.270***	1.000	9/9
	(0.101)	(0.050)	(0.054)	(0.066)		
Status at entry	+0.233**	−0.035	+0.279***	+0.169**	0.052	6/9
	(0.096)	(0.055)	(0.048)	(0.065)		
Excluding storefronts already operating in 2022Q2						
Lifetime	−0.052	−0.047	−0.005	+0.020	0.653	0/9
	(0.080)	(0.074)	(0.021)	(0.037)		
Predetermined	−0.066	−0.021	+0.100	+0.348***	0.998	9/9
	(0.102)	(0.032)	(0.060)	(0.077)		
At entry	−0.061	−0.035	+0.093**	+0.135**	0.797	4/9
	(0.095)	(0.055)	(0.038)	(0.050)		

Notes: Columns 1–2: 2022Q2 nonlegal-border coefficients for asinh operator counts by type. Columns 3–4: averages of $\hat{\beta}_{I,q} - \hat{\beta}_{C,q}$ over the early window (2022Q2–2023Q1) and the late window (2024Q2–2026Q2); positive values mean the gap is larger for independents. Column 5: two-one-sided-test p-value for H_0 : |late mean| $\geq$ |early mean|. Column 6: number of late quarters with an individually significant difference. Lifetime: chain if the operator runs $\geq$2 NM storefronts at any point in the sample. Predetermined: chain if the operator runs $\geq$2 storefronts in 2022Q2. At entry: a storefront is an established-operator entry if its operator already ran another NM storefront when it first sold; by construction no storefront operating in 2022Q2 is an established-operator entry, so the full-sample early difference under this classification is mechanical. The lower panel counts only storefronts that first report sales after 2022Q2; there, the window averages are the informative statistics. 209 ZIP codes; county-clustered standard errors in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

6.3 Who opened the new stores?

The comparisons above count operators present in each quarter, so they combine openings and closures. Table 12 counts the 426 openings after 2022Q2 in the main sample directly and classifies each by its operator's status at entry: an additional store of an established operator, the store of a single-store operator, or the first location of an operator that later opens more stores (this last split uses look-ahead). Nonlegal-border ZIP codes saw more openings of every kind. Per 100,000 residents, established operators opened 24.3 additional stores in nonlegal-border ZIP codes and 10.5 in always-legal-border ZIP codes; single-store operators opened 33.4 and 14.6; later-expanding operators opened 11.5 and 4.1 first locations. In Poisson regressions with a population control and quarter fixed effects, the opening rate in nonlegal-border ZIP codes is 2.15 times that in always-legal-border ZIP codes for established operators' additional stores, 2.38 times for single-store operators, and 2.52 times for first locations, although the last estimate is imprecise (95 percent interval 0.89–

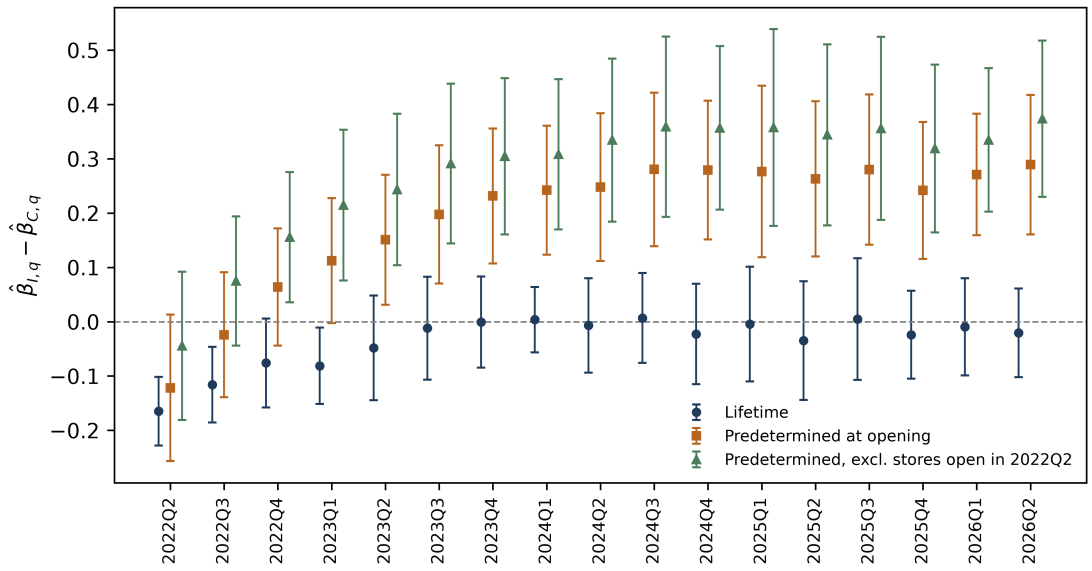


Fig. 6 Independent-minus-chain difference under alternative classifications

Notes: Quarterly $\hat{\beta}_{I,q} - \hat{\beta}_{C,q}$ with 95% confidence intervals from the regressions in Table 11: lifetime classification (all storefronts), classification at market opening (all storefronts), and classification at market opening excluding storefronts operating in 2022Q2. Positive values mean the nonlegal-border gap is larger for independents.

7.11); established operators' second stores, the step that turns an independent into a chain, open at 3.13 times the rate. The shares of the three kinds among openings are almost identical in the two groups, and a stacked Poisson regression does not reject equal proportional responses. Captive demand roughly doubled openings by every kind of operator and left the mix of new stores unchanged, which is P5.

6.4 What predicts early entry?

If the head start reflects inherited capacity, operators that were already set up to sell cannabis should also have opened their new stores sooner. Table 13 relates the timing of the 426 openings to two characteristics of the operator measured before adult-use sales began: whether it had a storefront with medical sales in April 2022, and how many storefronts it ran in 2022Q2. Operators with medical storefronts opened their new stores 3.2 quarters earlier (SE 0.9), against an average delay of 7.3 quarters, and were 49 percentage points more likely to open them within the first year (SE 18), where the average is 24 percent. These estimates hold operator size fixed. Size itself does not speed up entry: operators with more stores at opening were, if anything, less likely to open a new store within the first year (−0.18, SE 0.08). New stores in nonlegal-border ZIP codes did

Table 12: New storefronts by operator status at entry

	Entries per 100k res.		PPML	OLS, entries per 100k		Share of
	NL	LB	ZIP-qtr	unweighted	pop.-weighted	entrants
	(1)	(2)	(3)	(4)	(5)	(6)
Established operator, additional store	24.3	10.5	0.765**	0.485	0.732***	−0.014
	[127]	[23]	(0.309)	(0.673)	(0.259)	(0.044)
of which: operator's 2nd store	10.3	3.6	1.141***	−0.184	0.441***	0.034
	[54]	[8]	(0.402)	(0.615)	(0.142)	(0.039)
Single-store operator	33.4	14.6	0.867***	0.670*	1.168***	0.000
	[175]	[32]	(0.240)	(0.404)	(0.354)	(0.067)
First location of later-expanding operator	11.5	4.1	0.924*	0.353	0.383**	0.013
	[60]	[9]	(0.529)	(0.229)	(0.160)	(0.042)
All entries	69.2	29.2	0.841***	1.508	2.283***	
	[362]	[64]	(0.259)	(0.951)	(0.643)	

Notes: Entries are storefronts' first quarters with positive sales, excluding storefronts operating in 2022Q2; 426 entries in the 209 main-sample ZIP codes. Established operator: the operator already ran another NM storefront when the store first sold; "operator's 2nd store" is the second NM storefront of such an operator. Single-store operator: the operator never runs a second NM storefront in the sample. First location of later-expanding operator: the operator's first NM storefront, with at least one more opened later. Columns 1–2: entries per 100,000 residents over the sample, with counts in brackets. Column 3: Poisson coefficient on Nonlegal for ZIP-quarter entry counts with log population and quarter fixed effects; $\exp(\hat{\beta})$ is the ratio of opening rates. Columns 4–5: OLS on entries per 100,000 residents, unweighted and population-weighted. Column 6: linear probability model of the entrant's type on Nonlegal among entrants. County-clustered standard errors in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

not open earlier than new stores elsewhere once operator characteristics are held fixed. The same conclusion holds at the operator level with classifications that avoid look-ahead: when a chain is defined as an operator with two stores within four quarters of its first, the chain head start in the timing of first stores (1.6 quarters) disappears once medical incumbency is held fixed; with an eight-quarter window, medical incumbency accounts for about 60 percent of it.

The regression is estimated by ordinary least squares, and medical incumbency is an explanatory variable, not an instrument. It could not serve as one: it plausibly affects timing directly, through licenses, compliant premises, supply contracts, and familiarity with the regulator, which is exactly what inherited capacity means here. The coefficient is a conditional comparison, not the causal effect of a medical license. Legalization plays a narrower role. The Act set a single statewide date on which sales could begin everywhere, chosen by no firm and common to both groups of markets, so timing measured from that date is comparable across operators. Medical incumbency and operator size are measured at or before that date and cannot respond to later entry, although they are not independent of the expectation of legalization: the Act passed a year before sales began.

The evidence therefore narrows the interpretation of the chain lead and supports P3 and P5. Cross-border demand was first served by retailers already licensed and operating when adult-use sales began, most of them medical dispensaries of multi-store operators; once new entry is examined

Table 13: Predetermined operator characteristics and the timing of entry

	Quarters from opening to entry	Entered within first year
log(1 + operator storefronts in 2022Q2)	+0.753	−0.177**
	(0.657)	(0.083)
Operator was a medical incumbent	−3.200***	+0.487***
	(0.941)	(0.182)
Nonlegal-border ZIP	+0.051	−0.028
	(0.559)	(0.047)
log(1 + storefronts in 2022Q2)× Nonlegal	−0.173	+0.032
	(0.606)	(0.061)
Mean of outcome	7.34	0.24
Storefronts (entered after 2022Q2)	426	

Notes: Storefronts in the main-sample ZIP codes that first report sales after 2022Q2, estimated by OLS. Column 1: quarters between 2022Q2 and the storefront's first quarter of sales. Column 2: indicator for entering by 2023Q1. Operator storefronts in 2022Q2 and medical incumbency (the operator had at least one storefront with medical sales in April 2022) are measured before any post-legalization entry. Standard errors clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

separately, chains show no advantage, and prior medical experience predicts earlier openings. The source of the head start is consistent with research emphasizing pre-entry resources in the timing of entry (Helfat & Lieberman, 2002; King & Tucci, 2002; Lieberman & Montgomery, 1988), but it remains an interpretation: without license application dates, existing facilities and regulatory experience cannot be separated from financing, location choices, or faster regulatory processing.

7 An Estimated Model of Entry and Market Adjustment

The reduced-form evidence answers how large the border premium is and who supplied it first. A model is needed for two further questions: whether the pieces fit together quantitatively, and what the market would have looked like without inherited medical capacity, with faster entry, or with a cap on stores. The paper builds the simplest model that can address these questions with the available data and estimates it rather than calibrating it.

7.1 Choice of model

Before the model is specified, the mechanism it should contain is checked against the evidence. A natural candidate is a dynamic game in which incumbents earn temporary rents in high-exposure markets that entry gradually competes away. Section 5.4 does not support that mechanism: continuing stores earned no premium in exposed markets at the start, and their sales did not rise

relative to opening. The model therefore builds the adjustment around market expansion, inherited capacity, and the timing of entry, without a temporary-rent component. Nor does the paper estimate a fully dynamic oligopoly game (Aguirregabiria & Mira, 2007; Pakes, Ostrovsky, & Berry, 2007). Its additional parameters (sunk entry costs and continuation values in dollars) would require margins or costs that revenue data do not provide, the set of potential new operators is not observed, and the 16 post-opening quarters give little variation to separate strategic waiting from processing delays. The model is partially structural: it estimates how the flow of openings, exits, and revenue depends on demand and on the existing market, and it simulates counterfactual paths holding those estimated relationships fixed.

7.2 Specification

Markets are the 368 ZIP codes m, observed for $t = 1, \ldots, 16$ quarters after the market opened (2022Q3–2026Q2), with the 2022Q2 stock N_{m0} as the initial condition. Openings are of three types k: first locations of operators (F), additional stores of operators that held a medical incumbent (EM), and additional stores of other established operators (EN). The expected number of type-k openings is

$$\begin{aligned} E[n_{kmt}] = \exp(&\kappa_k + \beta_k \log L_m + \delta_k \log(1 + B_m) + \mu_k \log(1 + B_m^L) + \psi_k' W_m \\ &+ h_k(t) + \gamma_k \log(1 + N_{m,t-1}) + \omega_k \log(1 + N_{m0})), \end{aligned} \tag{4}$$

where L_m is resident population, B_m and B_m^L are nonlegal-state and legal-state residents within 75 miles (in thousands), W_m collects the predetermined controls of Section 5.5 and a tribal-land indicator, and $h_k(t) = \phi_{1k} \log t + \phi_{2k} t$ is the type-specific pipeline profile, common to all markets. Demand (L_m, B_m) determines how many entry projects are worth completing; $h_k(t)$ describes how fast they are completed after opening; and γ_k measures how the existing stock changes the flow of later openings. Competition lowers that flow, but persistent local attractiveness raises it. Following Wooldridge (2005), the initial stock $\log(1+N_{m0})$ enters as a proxy for such persistent heterogeneity. Stores exit with probability $\Lambda(\nu_0 + \nu_1 \log(1 + N_{mt}) + \nu_2 \log(1 + B_m) + \nu_3 \log L_m + \nu_4 \log t)$, and market revenue is

$$\log R_{mt} = \xi_m + a_t + \sigma \log N_{mt} + u_{mt}, \tag{5}$$

where ξ_m are market fixed effects and a_t quarter effects. In the simulations, ξ_m for ZIP codes that never host a store is predicted from their covariates. The stock evolves as $N_{mt} = N_{m,t-1} + \sum_k n_{kmt} - x_{mt}$, where x_{mt} is exits; the simulations use the expected-value recursion.

The model maps onto the framework. Through δ_k and β_k, captive demand raises the number of stores a market supports (P1); σ governs how additional stores translate into revenue and how incumbents' sales respond (P2, P4); the timing profiles of the EM and EN flows describe inherited capacity (P3); and the split of openings by type tests whether demand tilts the composition of entry (P5).

7.3 Identification and estimation

Table 14 summarizes what identifies each parameter and how firmly. The demand coefficients come from differences in openings across markets with different populations and exposure in the same quarter; they inherit Assumption 1. The pipeline profiles come from the timing of openings within the 16 quarters. The revenue elasticity σ comes from changes in a market's revenue as its number of stores changes, holding fixed the market and the quarter; it is biased upward if entry follows local demand shocks within a market, and it is compared with the independent store-level evidence of Section 5.4. The effect of the existing stock on later entry, γ_k, is not point identified. Pooled across markets, it absorbs persistent attractiveness and is biased upward even with the initial-condition proxy; with market fixed effects in a short panel, it is biased downward. Both estimates are therefore reported and used as bounds. Entry equations are estimated by Poisson pseudo-maximum likelihood, the exit equation by logit, and the revenue equation by least squares, all with county-clustered standard errors.

Table 15 reports the preferred specification. First locations of new operators rise with captive demand: each additional log point of nonlegal-state residents within 75 miles raises expected first-location openings by about 9 percent (0.085, SE 0.034), holding population, geography, and the existing stock fixed. Expansions by other established operators respond positively but imprecisely (0.055, SE 0.052). Expansions by medical-incumbent operators are directed toward larger markets (0.87 on log population) and, conditional on that, slightly away from exposed ones (-0.077, SE 0.035). The pipeline profiles differ by type in the way P3 predicts: openings by medical-incumbent operators are concentrated early and decline quickly, while expansions by other established opera-

Table 14: Parameters of the entry-and-adjustment model: interpretation and identification

Parameter	Interpretation	Identifying variation	Key assumption	Status
δ_k, β_k	Response of openings to captive and resident demand	Openings across markets with different exposure and population, same quarter	Conditional comparability (Assumption 1)	Estimated
ϕ_{1k}, ϕ_{2k}	Pipeline profile by opening type	Timing of openings after April 2022	Common across markets; parametric in t	Estimated
γ_k	Effect of the existing stock on later openings	Pooled: across markets; fixed effects: within market	Pooled biased up, fixed effects biased down	Bounded
ν	Exit hazard	Store-quarter exits	Logit; profile held at its last estimated value	Estimated
σ	Elasticity of market revenue to the store count	Within-market changes in stores and revenue	No within-market demand shocks timed with entry	Estimated (upper bound)
Cap c	Hypothetical stores per 1,000 residents	None (policy scenario)	Grandfathered stores; entry policy unchanged	Assumed

Notes: The model is estimated on all 368 New Mexico ZIP codes, 2022Q3–2026Q2. Revenue data do not identify prices, quantities, margins, or sunk costs in dollars; the model reports entry intensities and revenue elasticities only.

tors arrive later. The pooled coefficient on the existing stock is positive for all types (0.83 for first locations), whereas the within-market estimates are negative (-1.05 for first locations and -0.89 for expansions, SE 0.40 and 0.15); the truth lies between these bounds. Exits rise with the number of stores in the market (0.22, SE 0.07), consistent with crowding. The revenue elasticity is $\hat{\sigma} = 0.81$ (SE 0.10): an additional store raises market revenue, but by less than its own sales, so revenue per store falls with elasticity $\hat{\sigma} - 1 = -0.19$. This is close to the store-level association between incumbents' sales and nearby rivals (-0.24 within 2 kilometers), which is estimated from a different source of variation.

7.4 Validation

The model is validated by estimating it on the first eight post-opening quarters (2022Q3–2024Q2) and predicting 2024Q3–2026Q2 from the observed 2024Q2 stock, and it is compared with simpler specifications: S0 contains only population and the pipeline profile, S1 adds exposure and covariates, S2 adds the lagged stock, and S3 adds the initial-condition proxy. Table 16 and Figure 7 report the results. Out of sample, the preferred specification predicts the ZIP-level stock in 2026Q2 with a root-mean-squared error of 1.42 stores, against 1.62 for S0, and predicts the 2026Q2 stock within about 20 percent in each exposure group (251 predicted against 235 observed at the nonlegal border, 44 against 55 at the always-legal border, and 393 against 350 in the interior). In sample, the model

Table 15: Entry, exit, and revenue: estimates of the preferred specification

	Openings (Poisson)		
	First location of an operator	Expansion, medical incumbent operator	Expansion, other operator
Log resident population	+0.462***	+0.868***	+0.464***
	(0.072)	(0.136)	(0.100)
Log nonlegal-state residents within 75 mi, log(1 + B/1000)	+0.085**	−0.077**	+0.055
	(0.034)	(0.035)	(0.052)
Log legal-state residents within 75 mi	−0.045	−0.324***	−0.143***
	(0.035)	(0.106)	(0.050)
Log population density	+0.145***	+0.074	+0.090
	(0.047)	(0.067)	(0.065)
Distance to Albuquerque (100 mi)	+0.140	+0.314**	+0.082
	(0.116)	(0.150)	(0.203)
Distance to nearest city ≥50k (100 mi)	+0.458*	+0.634**	+0.487
	(0.261)	(0.321)	(0.341)
Retail establishments per 1,000, 2019 (asinh)	+0.299***	+0.403**	+0.436***
	(0.097)	(0.199)	(0.085)
Centroid on tribal land	−0.343	+0.416	−0.217
	(0.234)	(0.467)	(0.227)
Log(1 + operating stores), previous quarter	+0.833***	+0.277	+0.909***
	(0.188)	(0.326)	(0.168)
Log(1 + stores in 2022Q2), initial-condition proxy	−0.269*	+0.118	−0.424***
	(0.164)	(0.233)	(0.145)
Log quarters since opening	−0.376**	−0.209	+0.350
	(0.162)	(0.220)	(0.356)
Quarters since opening	−0.066**	−0.115***	−0.037
	(0.027)	(0.042)	(0.038)
Openings	558	160	210
ZIP-quarters	5,888 (368 ZIPs, 2022Q3–2026Q2)		
Exit (logit, store-quarters) and revenue			
Exit: log(1 + operating stores in ZIP)	+0.218***		
	(0.071)		
Exit: log nonlegal-state residents	−0.005		
	(0.016)		
Exit: log population	−0.073		
	(0.056)		
Exit: log quarters since opening	+0.791***		
	(0.075)		
Revenue: elasticity of market revenue to the store count, σ	+0.806***		
	(0.104)		

Notes: Poisson pseudo-maximum-likelihood estimates of equation (4) for 2022Q3–2026Q2 openings in the 368 New Mexico ZIP codes, specification S3; exit logit on store-quarters; revenue elasticity from equation (5) with ZIP and quarter fixed effects on the 1,745 ZIP-quarters with at least one store. County-clustered standard errors in parentheses. The within-ZIP Poisson estimates of the stock coefficient used as the alternative bound are −1.051 (SE 0.403) for first locations and −0.887 (SE 0.151) for expansions. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

reproduces the timing of expansions by medical-incumbent operators (mean opening quarter 5.75 predicted, 5.72 observed) and the composition of openings by type and exposure group. The model also has clear weaknesses. It underpredicts late openings (242 against 330) while overpredicting survival, and it overstates the late-sample growth of the border gap: the predicted ratio of revenue per resident between the two groups is 3.34, against 2.84 observed, and the predicted exposure gradient in store density is about twice the observed one. The simplest specification does better on these two moments, because the observed 2024Q2 stock already embeds the border gap. The analysis therefore emphasizes counterfactual comparisons that hold the model's biases roughly constant across scenarios and reports the bounds on γ_k throughout.

Table 16: Model validation and comparison with simpler specifications

	S0: population and time profile	S1: + exposure and covariates	S2: + lagged stock	S3: + initial stock (preferred)
Out-of-sample: estimated on 2022Q3–2024Q2, predicting 2024Q3–2026Q2				
RMSE of ZIP-level stock in 2026Q2	1.62	1.55	1.45	1.42
Openings, 2024Q3–2026Q2 (observed 330)	305	305	270	242
Stores in 2026Q2, nonlegal border (observed 235)	234	264	258	251
Stores in 2026Q2, always-legal border (observed 55)	60	49	46	44
Stores in 2026Q2, interior (observed 350)	456	430	409	393
Nonlegal/always-legal revenue per resident (observed 2.84)	2.78	3.17	3.31	3.34
Exposure gradient of 2026Q2 store density (observed 0.022)	0.023	0.047	0.047	0.041
In-sample path from the observed 2022Q2 stock				
RMSE of ZIP-level stock in 2026Q2	2.53	1.81	1.84	1.90

Notes: Out-of-sample rows: parameters estimated on 2022Q3–2024Q2 (exit hazard profile held at its last estimated value), simulated from the observed 2024Q2 stock to 2026Q2. In-sample row: full-sample parameters simulated from the observed 2022Q2 stock. RMSE: root-mean-squared error across the 368 ZIP codes. The exposure gradient is the coefficient of asinh stores per 1,000 residents in 2026Q2 on $\log(1 + B/1000)$, controlling for log population, across 368 ZIP codes. Stores are counted from first to last quarter with sales.

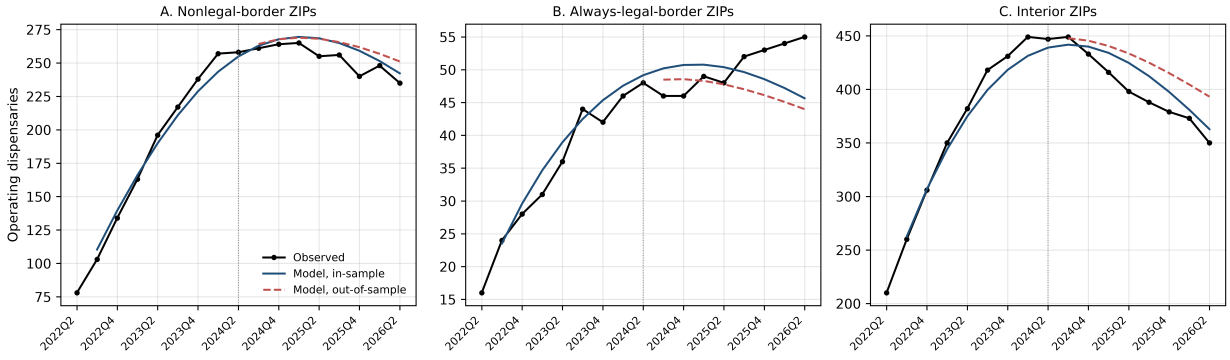


Fig. 7 Observed and predicted operating dispensaries by exposure group

Notes: Observed stock of operating dispensaries (first to last quarter with sales), the in-sample path of the preferred specification simulated from the observed 2022Q2 stock, and the out-of-sample path estimated on 2022Q3–2024Q2 and simulated from the observed 2024Q2 stock (vertical line).

8 Counterfactuals and Constrained Entry

Each scenario is simulated from 2022Q2 to 2026Q2 under both versions of the entry block, one in which the existing stock raises later openings (the pooled estimate) and one in which it lowers them (the within-market estimate), and compared with the model's own baseline path rather than with the data. Intervals are the 5th and 95th percentiles across 300 draws of all parameters from their estimated sampling distributions. Table 17 and Figure 8 report the results. The first two scenarios change inputs the model estimates directly; the third imposes a policy the data never observe and depends more heavily on the model's assumptions.

Removing the 226 medical-incumbent storefronts from the 2022Q2 stock lowers store-quarters in the first year by 64 percent if the existing stock attracts later entry and by 19 percent if it deters it. When expansions by medical-incumbent operators also lose their timing advantage, the first-year losses are 69 and 30 percent. The sign is the same under both bounds, and the effect is large: inherited capacity is what allowed the market to serve demand from the first quarter. The long-run effect is not identified. If existing stores attract entrants, the market never catches up and store-quarters over the whole sample fall by about half; if existing stores deter entrants, new operators fill most of the gap, openings rise by about 200, and total store-quarters fall by only 5 to 9 percent. The data cannot distinguish these cases, because they turn on γ_k. In both cases revenue per store changes by less than 3 percent, so inherited capacity affected how quickly the market reached its size, not how sales were split among stores.

Running the entry schedule one quarter faster, so that each opening occurs one quarter earlier, raises store-quarters in the first year by 11 to 17 percent and over the whole sample by 5 to 6 percent; two quarters faster raises them by 8 to 13 percent. Revenue rises almost in proportion and revenue per store falls slightly, as the revenue elasticity implies. Faster licensing therefore mainly brings the market's expansion forward rather than changing its eventual size.

Finally, each ZIP code is capped at c stores per 1,000 residents, with at least one store allowed and existing stores grandfathered, for $c \in \{0.10, 0.19, 0.30\}$; 0.19 is the average density of always-legal-border ZIP codes. A cap of 0.19 reduces store-quarters by 36 to 38 percent and model revenue by 32 to 34 percent, and it raises revenue per store by about 8 percent. It also changes how the border premium is absorbed: the share of the gap in revenue per resident between nonlegal- and

always-legal-border markets carried by store density falls from 0.77 to 0.40–0.48, so more of the premium appears as sales per store, as P6 predicts. These magnitudes rely on the estimated revenue elasticity and on the assumption that capped entrants simply do not open; they say nothing about prices, margins, or welfare, which the data cannot identify, and they are not an estimate of any existing regulatory regime.

Table 17: Counterfactual simulations

	A: stock raises later openings			B: stock lowers later openings		
Scenario (percent change vs. model baseline)	Store-quarters	Revenue	Rev./store	Store-quarters	Revenue	Rev./store
No inherited medical outlets	-49.0	-47.8	+0.9	-4.7	-0.1	+2.6
	[-55.6, -37.7]	[-58.3, -34.0]	[-12.5, +13.8]	[-5.8, -2.0]	[-1.3, +2.5]	[+1.7, +3.2]
No inherited outlets and no timing advantage	-57.6	-58.3	-2.6	-8.7	-5.1	+1.8
	[-62.7, -47.7]	[-66.8, -45.8]	[-18.7, +13.9]	[-10.2, -5.9]	[-6.9, -2.2]	[+0.9, +2.6]
Entry schedule one quarter faster	+4.7	+4.1	-0.7	+6.2	+5.6	-0.7
	[+4.0, +5.3]	[+3.3, +5.1]	[-1.3, -0.0]	[+6.0, +6.6]	[+4.8, +6.7]	[-1.6, +0.2]
Entry schedule two quarters faster	+8.3	+7.2	-1.2	+13.4	+11.9	-1.6
	[+6.9, +9.8]	[+5.8, +9.1]	[-2.4, -0.1]	[+13.0, +14.5]	[+10.1, +14.4]	[-3.6, +0.4]
Cap: 0.30 stores per 1,000 residents	-27.8	-23.6	+7.0	-25.6	-21.7	+6.3
	[-30.9, -25.9]	[-29.8, -18.8]	[-0.3, +13.7]	[-28.3, -24.0]	[-27.3, -18.1]	[+0.2, +13.1]
Cap: 0.19 stores per 1,000 residents	-37.7	-33.6	+8.5	-35.6	-31.5	+8.2
	[-40.7, -35.9]	[-41.0, -27.2]	[-1.8, +18.7]	[-38.5, -34.0]	[-38.2, -26.3]	[-1.1, +18.1]
Cap: 0.10 stores per 1,000 residents	-44.9	-42.5	+6.7	-42.9	-40.5	+6.5
	[-47.7, -43.0]	[-50.3, -35.2]	[-6.1, +19.3]	[-45.7, -41.3]	[-47.7, -34.2]	[-5.0, +19.1]
First year only (store-quarters 2022Q2–2023Q1; revenue 2022Q3–2023Q1)						
No inherited medical outlets	-63.8	-60.3		-19.1	-0.6	
	[-66.5, -57.6]	[-68.5, -48.1]		[-22.2, -12.6]	[-3.8, +6.1]	
No inherited outlets and no timing advantage	-69.2	-67.9		-30.3	-13.8	
	[-70.9, -65.6]	[-74.1, -60.4]		[-34.4, -22.9]	[-18.4, -6.4]	
Entry schedule one quarter faster	+11.4	+10.9		+17.0	+16.6	
	[+10.5, +12.9]	[+8.8, +13.5]		[+16.0, +19.1]	[+14.1, +20.3]	
Entry schedule two quarters faster	+21.0	+19.8		+35.5	+34.0	
	[+18.9, +24.1]	[+15.8, +24.8]		[+33.1, +40.3]	[+28.3, +42.7]	
Baseline (A): store-quarters 10,909; revenue $2.70 billion; density share of the NL–LB gap 0.77						
Baseline (B): store-quarters 10,494; revenue $2.63 billion; density share of the NL–LB gap 0.78						

Notes: Percent changes relative to the model's own baseline path, 2022Q2–2026Q2, for all 368 New Mexico ZIP codes. Version A uses the pooled estimate of the effect of the existing stock on later openings (specification S3); version B fixes that effect at the within-ZIP Poisson estimate and re-estimates the remaining entry parameters. Brackets: 5th and 95th percentiles across 300 draws of all parameters (seed 20261023). Revenue is model revenue from equation (5). Caps are per ZIP code, with at least one store and grandfathered existing stores.

8.1 A descriptive comparison with Colorado.

Colorado lets local governments prohibit or restrict recreational cannabis retail, and many have done so. A county-level version of the border comparison for New Mexico and Colorado over 2022Q2–2025Q2, with the border indicator interacted with an indicator for Colorado, points in the direction of P6 (Online Resource 1, Tables OR20–OR22). The density border premium is smaller in Colorado (interaction −0.93, SE 0.49), the revenue-per-store premium larger (0.47, SE 0.31), and the revenue-per-resident premium smaller (−0.46, SE 0.39); only the density interaction approaches conventional significance. The comparison does not identify the effect of entry restrictions. Col-

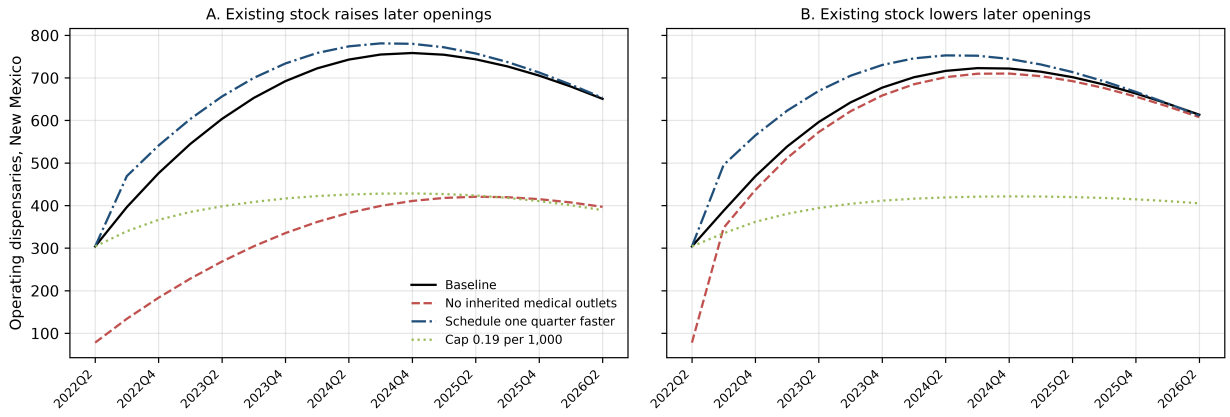


Fig. 8 Simulated stock of dispensaries under alternative scenarios

Notes: Expected stock of operating dispensaries in New Mexico under the model's baseline and three scenarios, for (a) the version in which the existing stock raises later openings and (b) the version in which it lowers them.

orado's market opened eight years earlier and was mature during the sample, its border counties face different neighbors and tourist flows, the binding local restrictions are not observed, and the analysis uses 74 counties. It is consistent with evidence that restricted entry raises incumbent market power in cannabis retail (Hollenbeck & Uetake, 2021) and with a working paper by Danna Thomas (2019) on license quotas in Washington, and with the broader evidence that entry regulation shapes local market structure (Schaumans & Verboven, 2008; Suzuki, 2013), but it remains descriptive.

9 Conclusion

A market that gains customers can serve them with the stores it has or with new ones. In New Mexico's cannabis market, which opened without a cap on licenses, captive demand from neighboring prohibition states was served mainly by new stores. Exposed markets host about twice as many dispensaries and generate about 2.4 times the revenue of comparable markets near legal states; roughly three-quarters of the revenue gap reflects store density; the premium is concentrated in adult-use sales and rises with the out-of-state population within driving distance; and stores that were already open earned no lasting premium. Controls for geography and an estimator that relies only on unexposed markets reduce the gap by up to half, so the magnitudes should be read as an upper range, but its pattern across margins is stable.

Who served the demand first depended on inherited capacity rather than organizational form.

Chains appear to move first only because the medical dispensaries that converted on the first day belonged mostly to multi-store operators; among stores opened later, chains and independents responded in proportion, and experience in the medical market, not the number of stores an operator ran, predicts early openings. An estimated model of entry and market adjustment reproduces these patterns and implies that inherited capacity mainly accelerated the market's start, that faster licensing would have brought the expansion forward, and that a cap on stores would have shifted the adjustment toward higher sales per store.

The findings bear on how new retail markets are opened. Where licensing does not cap the number of stores, a localized increase in demand is spread across more retailers; where a market opens to firms holding related licenses, those firms capture the first wave of demand. Three kinds of data would sharpen these conclusions: license application and approval records, which would separate preparation from regulatory delay; transaction data with prices and quantities, which would turn revenue elasticities into statements about consumers and margins; and within-state variation in local entry rules, which would allow a credible test of the effect of entry restrictions.

Statements and Declarations

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Competing interests. The author declares no competing interests.

Data availability. The cannabis sales records are public monthly workbooks published by the New Mexico Cannabis Control Division (New Mexico Cannabis Control Division, 2026). ZIP-code populations and coordinates come from the SimpleMaps U.S. ZIP Codes Database, Basic version, used under a Creative Commons Attribution 4.0 license (SimpleMaps, 2026); census blocks, tribal-area boundaries, state boundaries, and ZIP Codes Business Patterns are public U.S. Census Bureau products (U.S. Census Bureau, 2021, 2024, 2025); road travel times were computed with the public Open Source Routing Machine service on OpenStreetMap data (Project OSRM, 2026); Colorado data are public records of the Colorado Department of Revenue (Colorado Department of Revenue, 2026; Colorado Department of Revenue, Marijuana Enforcement Division, 2026). A replication package with all code, the constructed panels, and the shareable raw inputs accompanies this submission and will be deposited in a public repository.

Use of generative AI. Generative AI is only used for grammar adjustments. All the outputs are verified and the author is responsible for all the outputs, sentences and results in the paper.

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A Proofs

Proof of Proposition 1. The objective $W(p; A) = R(p) - K(p; A)$ is continuous on the compact set $[0, 1]$, so a maximizer exists. Because $W_{pp} = R_{pp} - K_{pp} < 0$, W is strictly concave in p and the maximizer $p^*(A)$ is unique. Since $W_{pA} = -K_{pA} > 0$, W has strictly increasing differences in (p, A), and by Topkis's monotonicity theorem the maximizer is weakly increasing in A. If $p^*(A) \in (0, 1)$, it satisfies $R_p(p^*) = K_p(p^*; A)$, and the implicit function theorem gives $dp^*/dA = K_{pA}/(R_{pp} - K_{pp})$, which is positive because numerator and denominator are both negative. For the final claim, suppose $A_H > A_L$, $p^*_L < 1$, $p^*_H > 0$, and $p^*_H = p^*_L \equiv \bar{p}$. If $\bar{p} \in (0, 1)$, the first-order conditions give $K_p(\bar{p}; A_L) = R_p(\bar{p}) = K_p(\bar{p}; A_H)$, contradicting $K_{pA} < 0$. If $\bar{p} = 0$, then $p^*_H = 0$, and if $\bar{p} = 1$, then $p^*_L = 1$, both contradictions. Hence $p^*_H > p^*_L$. □

Proof of Proposition 2. Subtracting the expression for $k = L$ from that for $k = H$ gives $\beta_H(t) - \beta_L(t) = \beta^\infty(\phi_H - \phi_L)e^{-\lambda t}$. Proposition 1 and $\phi' > 0$ give $\phi_H \geq \phi_L$, with strict inequality if $p^*_H > p^*_L$. The derivative with respect to t is $-\lambda\beta^\infty(\phi_H - \phi_L)e^{-\lambda t}$, which is non-positive and strictly negative when $\beta^\infty > 0$ and $\phi_H > \phi_L$. □

Fit of the common-pipeline path. Fitting $\beta_k(t)$ by weighted nonlinear least squares to the 34 quarterly coefficients of Table 10 gives $\hat{\beta}^\infty = 0.36$, $\hat{\lambda} = 0.41$ per quarter (a half-life of $\ln 2/\hat{\lambda} \approx 1.7$ quarters), $\hat{\phi}_C = 0.57$, and $\hat{\phi}_I = 0.09$. The implied chain lead, $\hat{\beta}^\infty(\hat{\phi}_C - \hat{\phi}_I)e^{-\hat{\lambda}t}$, starts at about 0.17 and falls to about 0.015 by 2023Q4. Under the classification at market opening the initial shares cannot be distinguished ($\hat{\phi}_C - \hat{\phi}_I = 0.45$, SE 0.75), and under the status-at-entry classification the ordering reverses mechanically, because every store operating at opening counts as a new-operator entry (Online Resource 1, Table OR16 and Figure OR4).